

Numerical Simulation of Convection-Diffusion Equation

Abstract

The convection-diffusion equations are an important type of motion equations, often used to describe the transmission and diffusion processes of some physical quantities such as temperature and solute. Therefore, it is of great significance to construct a fast, accurate and stable numerical algorithm for the convection-diffusion equations. In practical applications, the equations often appear to be convection-dominated. In this case, traditional difference methods are prone to numerical oscillations and make the calculation results poor. At present, the research on the numerical solution of the convection-dominated problems has become one of the hot spots in the field of numerical calculation, and the characteristic difference method is one of the most commonly used existing algorithms.

This article first introduces the one-dimensional characteristic difference scheme in detail. We find that the traditional characteristic difference scheme is prone to large numerical diffusion when the convection term coefficients are large. In order to eliminate this numerical phenomenon, on the basis of the characteristic difference scheme, we have improved the discreteness of the derivative of the characteristic direction. When the coefficients of the convection term are large, new interpolation points are selected along the characteristic line and bilinear interpolation is used; when the coefficients of the convection term are small, the interpolation points and the interpolation method of the characteristic difference scheme are still used. This paper first constructs an improved characteristic difference scheme for one-dimensional linear problems, and gives its error analysis under constant coefficients. We also extended the improved characteristic difference scheme to one-dimensional nonlinear and two-dimensional linear problems. At the same time, we provide numerical experiments to verify the effectiveness of the improved characteristic difference scheme for the above problems. It can be seen from the error analysis and numerical experiments that the improved characteristic difference scheme constructed in

this paper has two main advantages: One is that the scheme solves the problem of large numerical diffusion of the characteristic difference scheme when the convection term coefficients are large, and has high numerical accuracy. The second is that compared to other improved scheme for the characteristic difference scheme, this scheme takes full advantage of the characteristic difference scheme when the coefficients of convection term are small. Therefore, the improved characteristic difference scheme we constructed has broader range of applications and higher numerical accuracy than the existing characteristic difference scheme.

Key Words: Convection-Diffusion Equation; Convection Dominance Problem; Characteristic Difference Scheme; Improved Characteristic Difference Scheme

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Chapter 1 Introduction

1.1 Background

In real life, a large number of physics and engineering problems can be solved by abstracting them into partial differential problems. For some simple partial differential problems, we can obtain their analytical solutions by methods such as Laplace transform. However, for more complex partial differential problems, we often cannot obtain their analytical solutions directly, so it is very necessary to numerically solve partial differential equations.

The convection-diffusion equations are a very important type of partial differential equations. It is often used to describe the convection and diffusion process of some physical quantities such as temperature and solute. It plays an important role in many fields such as chemical reactions, fluid mechanics, and electromagnetic field theory. Therefore, how to construct the numerical solution of the convection-diffusion equations has always been an important and popular research question. The difficulty of the numerical solution is to deal with a type of equations whose convection effect is far greater than the diffusion effect, the so-called convection-dominant diffusion equations. Using the usual difference methods to deal with this type of equations is prone to numerical oscillations, which makes the methods invalid. In the 1980s, Douglas and Russell et al. proposed a type of characteristic difference technique[1], solving the problem well. The research in this paper is based on the characteristic difference technique. The main contents of each chapter in the article are as follows: In Chapter 1, several numerical solutions for solving the convection-diffusion equation are briefly introduced and their corresponding advantages and disadvantages are discussed; In Chapter 2, the one-dimensional characteristic difference scheme is introduced in detail, and the scheme is further extended to two-dimensional scheme; In the Chapter 3, problems caused by the characteristic difference scheme are discussed, and one-dimensional linear and nonlinear improved characteristic difference schemes are constructed based on the characteristic difference scheme. Meanwhile the error analysis and numerical experiments of improved characteristic difference scheme are given. In Chapter 4, the

improved characteristic difference scheme is further extended to two-dimensional scheme, and two numerical experiments are given at the same time.

1.2 Literature Review

As an important type of motion equations, numerous researches have been down on its numerical solution. The traditional finite difference methods for dealing with the convection-diffusion equations include explicit scheme, implicit scheme, Crank-Nicolson scheme, etc. The explicit scheme has the characteristics of simple and small calculation, but its accuracy is poor, and it has strict convergence conditions, so it is less practical. The implicit scheme has good stability, but only first-order accuracy in the time direction. Naturally, we use the central difference method in the time direction to improve its accuracy. The Crank-Nicolson scheme is one of the common methods. Its accuracy is higher than the implicit scheme, and its stability is generally good. However, the coefficient matrix of the scheme is not always diagonally dominant, which has a great impact on its stability. In order to overcome the problem that the coefficient matrix of the central difference scheme is not always diagonally dominant, a corresponding penalty method and its improvement methods have been developed [3].

However, when dealing with a type of so-called convection-dominated diffusion equations, the above-mentioned traditional methods may produce serious numerical oscillations and thus fail. The characteristic difference technique proposed by Douglas and Russell et al. solves this problem well, and they combined the technique with the finite difference method and the finite element method to construct the characteristic finite difference scheme and the characteristic finite element scheme. At the same time, the error order of the two schemes are theoretically analyzed [1]. According to the characteristic finite difference scheme, the convection-diffusion equation is solved by constructing the characteristic directional derivative and discretizing it along the characteristic direction. This method solves the numerical oscillation caused by convection dominance, but in some cases, the effect of this method may still be poor [4]. In fact, the final effect of this method depends to a large extent on interpolation methods and the choice of interpolation points. Specifically, general linear interpolation can often avoid numerical oscillations, but there

will be greater numerical diffusion; although the quadratic interpolation can reduce the numerical diffusion, it may produce serious numerical oscillations; while the higher-order interpolation method has better results, but often the calculation amount is too large [5]. At present, a lot of researches on characteristics difference scheme has been completed. Lu Jinfu et al. [6] used diffusion correction on the diffusion term coefficients of the equation, constructed a new convection diffusion equation compatible with the original equation, and then performed linear interpolation to avoid numerical oscillations and reduce numerical diffusion. At the same time, in the paper [7], they constructed the so-called monotonic quadratic interpolation by combining the first and second Lagrange interpolations, which also got good numerical results. In paper [5], by combining the characteristic difference scheme and compact difference scheme, a new scheme is constructed. The results show that the scheme is stable and its effect is better than the traditional characteristic difference method using linear interpolation. Qin Xinqiang, Li Qiufang, etc. [2,4,8] construct a new characteristic difference scheme by selecting different points along the characteristic direction. The experimental results in their articles show that their scheme has good stability for different time and space steps.

In addition, the upwind difference scheme is another common numerical solution scheme for the convection-diffusion equations. The scheme fully considers the upwind effect of the convection term [9,10], so it has good stability. But the traditional upwind difference scheme has only first-order accuracy. On the basis of this scheme, Chen Guoqian and others have done further work to improve the accuracy of the traditional upwind difference scheme [11,12]. Other methods to numerically solve convection-diffusion equations include using non-uniform grids to construct difference equations [13,14] and deducing some convergence schemes that can theoretically reach any order of precision through the analysis of the error [15,16].

1.3 Symbol Introduction

Convention: The one-dimensional or two-dimensional grid divisions appearing in the article are all equidistant grid divisions, and the points where the grid lines intersect are called nodes.

The main symbol appearing in the article are as follows:

Table 1-1

Δt	Time Step
h	Space Step
$u(x_l, t^n), U_l^n$	The value of analytical solution u at the node (x_l, t^n) .
$u(x_l, y_i, t^n), U_{l,i}^n$	The value of analytical solution u at the node (x_l, y_i, t^n) .
u_l^n	The value of numerical solution u at the node (x_l, t^n) .
$u_{l,i}^n$	The value of numerical solution u at the node. (x_l, y_i, t^n) .
$[x]$	The largest integer not exceeding the real number x

Define the following norm in the domain of function u ,

$$\|u\|_{\infty} = \max_{x,t} |u(x,t)|, \quad \|u\|_{m,\infty} = \|u\|_{\infty} + \sum_{i=1}^m \left\| \frac{\partial^i u}{\partial x^i} \right\|_{\infty},$$

$$\|u^n\|_{\infty} = \max_x |u(x, t^n)|, \quad \|u^n\|_{m,\infty} = \|u^n\|_{\infty} + \sum_{i=1}^m \left\| \frac{\partial^i u^n}{\partial x^i} \right\|_{\infty}.$$

Chapter 2 Characteristic Difference Scheme

2.1 One-dimensional Characteristic Difference Scheme

2.1.1 Linear Case

Let us consider the following one-dimensional linear convection-dominated convection-diffusion problem

$$\begin{cases} \frac{\partial u}{\partial t} + b(x) \frac{\partial u}{\partial x} - \frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right) = f(x, t), & x \in (0, L), t \in (0, T], \\ u(x, 0) = u_0(x), & x \in (0, L), \\ u(0, t) = g_1(t), u(L, t) = g_2(t), & t \in [0, T], \end{cases} \quad (2-1-1)$$

where $a(x) \geq a_0 > 0$, $\min_x |b(x)| \gg \max_x a(x)$ and $u(x, t)$ is smooth in its domain.

Set

$$\varphi(x) = \sqrt{1 + b(x)^2}, \quad (2-1-2)$$

and use $\tau = \tau(x)$ to denote the characteristic direction accompanying the operator $u_t + b(x)u_x$, then we have

$$\tau = \tau(x) = \left(\frac{b(x)}{\varphi(x)}, \frac{1}{\varphi(x)} \right). \quad (2-1-3)$$

Taking the directional derivative along the above characteristic direction $\tau(x)$, then we can obtain the following operator

$$\frac{\partial}{\partial \tau(x)} = \frac{b(x)}{\varphi(x)} \frac{\partial}{\partial x} + \frac{1}{\varphi(x)} \frac{\partial}{\partial t}. \quad (2-1-4)$$

By bringing equation (2-1-4) back to equation (2-1-1), we can get the following expression equivalent to equation (2-1-1)

$$\begin{cases} \varphi(x) \frac{\partial u}{\partial \tau(x)} - \frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right) = f(x, t), & x \in (0, L), t \in (0, T], \\ u(x, 0) = u_0(x), & x \in (0, L), \\ u(0, t) = g_1(t), u(L, t) = g_2(t), & t \in [0, T]. \end{cases} \quad (2-1-5)$$

Next, we will derive the specific difference scheme of the above equation (2-1-5). Firstly, the space is divided uniformly, specifically: $x \in [0, L]$ is uniformly divided into p parts, and $h = L/p$; $t \in [0, T]$ is divided evenly into m parts, and $\Delta t = T/m$. At the same time, (x_l, t^n) is used to represent the node on the grid, where $0 \leq l \leq p$, $0 \leq n \leq m$, and l, n are integers.

Let us consider the case when $b(x) > 0$, because when $b(x) < 0$, the difference

scheme can be constructed in a similar way. Consider the characteristic line starting from node (x_l, t^n) , let $j = j_l = \left\lceil \frac{\Delta t b(x_l)}{h} \right\rceil$, then when $l > j$, the characteristic line and the grid line will intersect at point A (Figure 2-1), denote it as $(\bar{x}_l, t^{n-1})$; when $l \leq j$, the characteristic line will intersect the leftmost grid line at point C (Figure 2-2), denote it as $(x_0, \bar{t}^n)$. Next, we will treat the two cases separately.

Case 1

When $l > j$, the characteristic line and the grid line will intersect at point A (Figure 2-1), denote it as $(\bar{x}_l, t^{n-1})$. Let r_l^n be the error generated by approximating $\varphi(x) \frac{\partial u}{\partial \tau(x)}$ at the node (x_l, t^n) , then the approximation of $\varphi(x) \frac{\partial u}{\partial \tau(x)}$ at the node (x_l, t^n) can be expressed as

$$\begin{aligned} \left(\varphi(x) \frac{\partial u}{\partial \tau(x)} \right)_l^n &= \varphi(x_l) \frac{u(x_l, t^n) - u(\bar{x}_l, t^{n-1})}{|AB|} + r_l^n \\ &= \frac{1}{\Delta t} (u(x_l, t^n) - u(\bar{x}_l, t^{n-1})) + r_l^n, \end{aligned} \quad (2-1-6)$$

where $1 \leq l \leq p-1, l > j, 1 \leq n \leq m$ and $\bar{x}_l = x_l - \Delta t b(x_l)$.

The approximation of $\left[\frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right) \right]_l^n$ can be given by the method of central difference quotient. Let $\tilde{r}_l^n$ be the error generated by approximating $\frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right)$ at the node (x_l, t^n) . Then specifically we can get

$$\begin{aligned} \left[\frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right) \right]_l^n &= \delta_{\bar{x}} (a(x) \delta_x u(x, t^n))_l + \tilde{r}_l^n \\ &= \frac{1}{h} \left[a \left(x_{l+\frac{1}{2}} \right) \delta_{x+\frac{1}{2}} u(x_l, t^n) - a \left(x_{l-\frac{1}{2}} \right) \delta_{x-\frac{1}{2}} u(x_l, t^n) \right] + \tilde{r}_l^n \\ &= \frac{1}{h^2} \left[a \left(x_{l+\frac{1}{2}} \right) (u(x_{l+1}, t^n) - u(x_l, t^n)) - a \left(x_{l-\frac{1}{2}} \right) (u(x_l, t^n) - u(x_{l-1}, t^n)) \right] + \tilde{r}_l^n, \end{aligned} \quad (2-1-7)$$

where $a(x_{l+1/2}) = a((l+1/2)h)$, $a(x_{l-1/2}) = a((l-1/2)h)$.

By putting equations (2-1-6) and (2-1-7) back into equation (2-1-5), we can get

$$\begin{cases} \frac{1}{\Delta t} (u(x_l, t^n) - u(\bar{x}_l, t^{n-1})) - \delta_{\bar{x}} (a(x) \delta_x u(x, t^n))_l = f(x_l, t^n) + R_l^n, \\ \quad 1 \leq l \leq p-1 \text{ and } l > j, 1 \leq n \leq m, \\ u(x_l, 0) = u_0(x_l), \quad 1 \leq l \leq p-1 \text{ and } l > j, \\ u(0, t^n) = g_1(t^n), \quad u(L, t^n) = g_2(t^n), \quad 0 \leq n \leq m, \end{cases} \quad (2-1-8)$$

where $R_l^n = \tilde{r}_l^n - r_l^n$.

By ignoring R_l^n , and using $u_l^n, \bar{u}_l^{n-1}, f_l^n$ to replace $u(x_l, t^n), u(\bar{x}_l, t^{n-1}), f(x_l, t^n)$ respectively, we can get the characteristic difference scheme in Case 1

$$\begin{cases} \frac{1}{\Delta t}(u_l^n - \bar{u}_l^{n-1}) - \delta_{\bar{x}}(a(x)\delta_x u^n)_l = f_l^n, & 1 \leq l \leq p-1 \text{ and } l > j, 1 \leq n \leq m, \\ u_l^0 = u_0(x_l), & 1 \leq l \leq p-1 \text{ and } l > j, \\ u_0^n = g_1(t^n), u_p^n = g_2(t^n), & 0 \leq n \leq m. \end{cases} \quad (2-1-9)$$

Next, we shall deal with the approximation of $\bar{u}_l^{n-1}$. In paper [1], the author mentioned two interpolation methods about the approximation methods of $\bar{u}_l^{n-1}$, namely, the first and second Lagrange interpolation, and pointed out that when dealing with convection-dominant problems, it is better to use the first-order interpolation approximation. Thus, for the approximation of $\bar{u}_l^{n-1}$, we shall use $u_{l-j}^{n-1}, u_{l-j-1}^{n-1}$ to perform the interpolation, specifically

$$\bar{u}_l^{n-1} \approx \left(\frac{\Delta t b(x_l)}{h} - j\right) u_{l-j-1}^{n-1} + \left(1 + j - \frac{\Delta t b(x_l)}{h}\right) u_{l-j}^{n-1}. \quad (2-1-10)$$

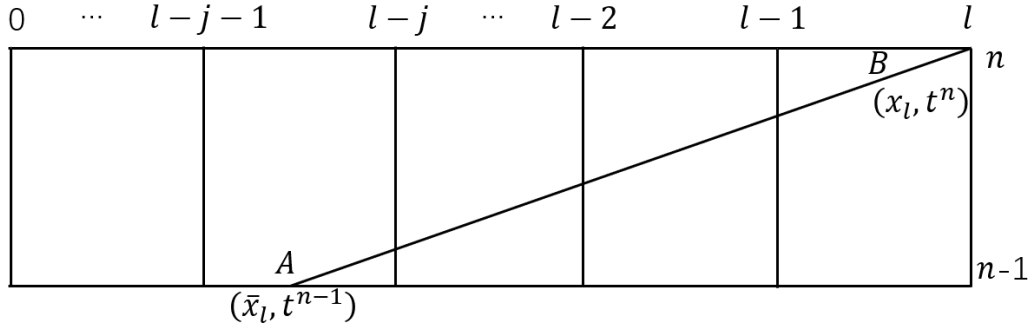


Figure 2-1

Case 2

When $l \leq j$, the characteristic line and the grid line will intersect at point C (Figure 2-2), denote it as $(x_0, \bar{t}^n)$. Let $\dot{r}_l^n$ be the error generated by approximating $\varphi(x) \frac{\partial u}{\partial \tau(x)}$ at the node (x_l, t^n) , then the approximation of $\varphi(x) \frac{\partial u}{\partial \tau(x)}$ at the node (x_l, t^n) can be expressed as

$$\begin{aligned} \left(\varphi(x) \frac{\partial u}{\partial \tau(x)}\right)_l^n &= \varphi(x_l) \frac{u(x_l, t^n) - u(x_0, \bar{t}^n)}{|CD|} + \dot{r}_l^n \\ &= \frac{b(x_l)}{lh} (u(x_l, t^n) - u(x_0, \bar{t}^n)) + \dot{r}_l^n, \end{aligned} \quad (2-1-11)$$

where $1 \leq l \leq p-1, l \leq j, 1 \leq n \leq m$ and $\bar{t}^n = t^n - \frac{lh}{b(x_l)}$.

The processing process and results of $\left[\frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right) \right]_l^n$ are consistent with (2-1-7).

Accodring to the approximation of $\left(\varphi(x) \frac{\partial u}{\partial \tau(x)} \right)_l^n$ and $\left[\frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right) \right]_l^n$, we can get

$$\begin{cases} \frac{b(x_l)}{lh} (u(x_l, t^n) - u(x_0, \bar{t}^n)) - \delta_{\bar{x}}(a(x) \delta_x u(x, t^n))_l = f(x_l, t^n) + \dot{R}_l^n, \\ 1 \leq l \leq p-1 \text{ and } l \leq j, 1 \leq n \leq m, \\ u(x_l, 0) = u_0(x_l), \quad 1 \leq l \leq p-1 \text{ and } l \leq j, \\ u(0, t^n) = g_1(t^n), \quad u(L, t^n) = g_2(t^n), \quad 0 \leq n \leq m, \end{cases} \quad (2-1-12)$$

where $\dot{R}_l^n = \tilde{r}_l^n - r_l^n$. Similar to the Case 1: By ignoring $\dot{R}_l^n$, and using $u_l^n, \tilde{u}_0^n, f_l^n$ to replace $u(x_l, t^n), u(x_0, \bar{t}^n), f(x_l, t^n)$ respectively, we can get the characteristic difference scheme in Case 2

$$\begin{cases} \frac{b(x_l)}{lh} (u_l^n - \tilde{u}_0^n) - \delta_{\bar{x}}(a(x) \delta_x u^n)_l = f_l^n, \quad 1 \leq l \leq p-1 \text{ and } l \leq j, 1 \leq n \leq m, \\ u_l^0 = u_0(x_l), \quad 1 \leq l \leq p-1 \text{ and } l \leq j, \\ u_0^n = g_1(t^n), \quad u_p^n = g_2(t^n), \quad 0 \leq n \leq m. \end{cases} \quad (2-1-13)$$

Similarly, the approximation value of $\tilde{u}_0^n$ is given by the interpolation of u_0^n and u_0^{n-1} , then we have

$$\tilde{u}_0^n \approx \left(1 - \frac{lh}{\Delta t b(x_l)} \right) u_0^n + \left(\frac{lh}{\Delta t b(x_l)} \right) u_0^{n-1}. \quad (2-1-14)$$

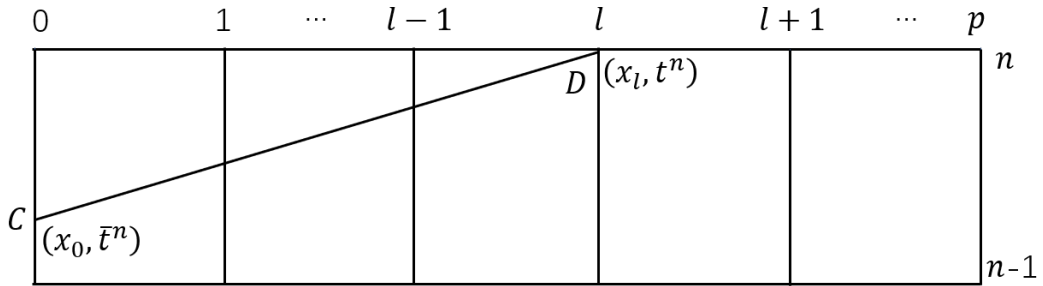


Figure 2-2

Combining Case 1 and Case 2, and let $s = s_l = \text{sign}(l - j(l))$, we can write the characteristic difference scheme as the following expression

$$\begin{cases} \frac{s(1+s)}{2\Delta t} (u_l^n - \bar{u}_l^{n-1}) + \left(1 - \frac{s(1+s)}{2} \right) \frac{b(x_l)}{lh} (u_l^n - \tilde{u}_0^n) - \delta_{\bar{x}}(a \delta_x u^n)_l = f_l^n, \\ 1 \leq l \leq p-1, 1 \leq n \leq m, \\ u_l^0 = u_0(x_l), \quad 1 \leq l \leq p-1, \\ u_0^n = g_1(t^n), \quad u_p^n = g_2(t^n), \quad 0 \leq n \leq m. \end{cases} \quad (2-1-15)$$

2.1.2 Non-linear Case

The characteristic difference scheme can be further extended to the following one-dimensional nonlinear problems

$$\begin{cases} c(x,u) \frac{\partial u}{\partial t} + b(x,u) \frac{\partial u}{\partial x} - \frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right) = f(x,t), & (x,t) \in (0,L) \times (0,T), \\ u(x,0) = u_0(x), & x \in (0,L), \\ u(0,t) = g_1(t), \quad u(L,t) = g_2(t), & t \in [0,T], \end{cases} \quad (2-1-16)$$

where $a(x) \geq a_0 > 0$, $c(x,u) \geq c_0 > 0$, $\min_{x,u} |b(x,u)| \gg \max_x a(x)$, and $u(x,t)$ is smooth in its domain.

We still consider the case when $b(x,u) > 0$, similarly when $b(x,u) < 0$. First let us replace u of $b(x,u), c(x,u)$ with u_l^{n-1} . For a fixed time layer t^n , u_l^{n-1} is known, so after the above replacement, we can use exactly the same method as the linear case to obtain the nonlinear characteristic difference scheme. Let $j = j_{l,n} = \left\lceil \frac{\Delta t b(x_l, u_l^{n-1})}{c(x_l, u_l^{n-1})h} \right\rceil$, then the nonlinear characteristic difference scheme in Case 1 can be derived as

$$\begin{cases} \frac{c(x_l, u_l^{n-1})}{\Delta t} (u_l^n - \bar{u}_l^{n-1}) - \delta_{\bar{x}}(a(x) \delta_x u^n)_l = f_l^n, & 1 \leq l \leq p-1 \text{ and } l > j, 1 \leq n \leq m, \\ u_l^0 = u_0(x_l), & 1 \leq l \leq p-1 \text{ and } l > j, \\ u_0^n = g_1(t^n), \quad u_p^n = g_2(t^n), & 0 \leq n \leq m, \end{cases} \quad (2-1-17)$$

where

$$\bar{u}_l^{n-1} \approx \left(\frac{\Delta t b(x_l, u_l^{n-1})}{c(x_l, u_l^{n-1})h} - j \right) u_{l-j-1}^{n-1} + \left(1 + j - \frac{\Delta t b(x_l, u_l^{n-1})}{c(x_l, u_l^{n-1})h} \right) u_{l-j}^{n-1}. \quad (2-1-18)$$

Similarly, the nonlinear characteristic difference scheme in Case 2 can be derived as

$$\begin{cases} \frac{b(x_l, u_l^{n-1})}{lh} (u_l^n - \tilde{u}_0^n) - \delta_{\bar{x}}(a(x) \delta_x u^n)_l = f_l^n, & 1 \leq l \leq p-1 \text{ and } l \leq j, 1 \leq n \leq m, \\ u_l^0 = u_0(x_l), & 1 \leq l \leq p-1 \text{ and } l \leq j, \\ u_0^n = g_1(t^n), \quad u_p^n = g_2(t^n), & 0 \leq n \leq m, \end{cases} \quad (2-1-19)$$

where

$$\tilde{u}_0^n \approx \left(1 - \frac{lh c(x_l, u_l^{n-1})}{\Delta t b(x_l, u_l^{n-1})} \right) u_0^n + \left(\frac{lh c(x_l, u_l^{n-1})}{\Delta t b(x_l, u_l^{n-1})} \right) u_0^{n-1}. \quad (2-1-20)$$

Set $s = s_{l,n} = \text{sign}(l-j) = \text{sign} \left(l - \left\lceil \frac{\Delta t b(x_l, u_l^{n-1})}{c(x_l, u_l^{n-1})h} \right\rceil \right)$ and combine Case 1 and Case 2, then, for the characteristic difference scheme of the nonlinear case, the following expression can be written

$$\begin{cases} \frac{s(1+s)c(x_l, u_l^{n-1})}{2\Delta t}(u_l^n - \bar{u}_l^{n-1}) + \left(1 - \frac{s(1+s)}{2}\right) \frac{b(x_l, u_l^{n-1})}{lh}(u_l^n - \bar{u}_0^n) - \delta_{\bar{x}}(a(x)\delta_x u^n)_l = f_l^n, \\ u_l^0 = u_0(x_l), \quad 1 \leq l \leq p-1, \\ u_0^n = g_1(t^n), \quad u_p^n = g_2(t^n), \quad 0 \leq n \leq m. \end{cases} \quad (2-1-21)$$

2.2 Error Analysis

In paper [1], the author gives the error analysis process and results of the characteristic difference scheme in one-dimensional case. In this article, in order to compare with the error analysis results of the improved characteristic difference scheme later, we give a error analysis of the one-dimensional constant coefficient characteristic difference scheme. Compared with the error analysis in the paper [1], we retain the coefficients as much as possible in the error analysis process, and give detailed proofs of omitted part in paper [1].

Let us consider the case when $l > j$. According to (2-1-6), we have

$$r_l^n = \varphi(x_l) \left(\frac{\partial u}{\partial \tau(x)} \right)_l^n - \varphi(x_l) \frac{u(x_l, t^n) - u(\bar{x}_l, t^{n-1})}{|AB|}. \quad (2-2-1)$$

Use $(x(\tau), t(\tau))$ to represent the coordinate on the characteristic line, so the Taylor expansion of $u(\bar{x}_l, t^{n-1})$ along the characteristic line τ can be obtained as follows

$$u(\bar{x}_l, t^{n-1}) = u(x_l, t^n) - |AB| \left(\frac{\partial u}{\partial \tau(x)} \right)_l^n - \int_{(x_l, t^n)}^{(\bar{x}_l, t^{n-1})} \sqrt{(x(\tau) - \bar{x}_l)^2 + (t(\tau) - t^{n-1})^2} \frac{\partial^2 u}{\partial \tau^2} d\tau. \quad (2-2-2)$$

After processing the above expression, we can get

$$r_l^n = \frac{1+b^2}{2} \left(\frac{\partial^2 u}{\partial \tau^2} \right)_{l^*}^{n*} \Delta t, \quad (2-2-3)$$

where $\left(\frac{\partial^2 u}{\partial \tau^2} \right)_{l^*}^{n*}$ is the value of the second-order tangent derivative of u along the line segment AB at a certain point on AB .

According to (2-1-7) we have

$$\tilde{r}_l^n = \left[\frac{\partial}{\partial x} \left(a \frac{\partial u}{\partial x} \right) \right]_l^n - \delta_{\bar{x}}(a \delta_x u(x, t^n))_l. \quad (2-2-4)$$

By performing Taylor expansion on $u(x_{l+1}, t^n), u(x_{l-1}, t^n)$ respectively at node (x_l, t^n) , and putting the result into equation (2-2-4), we can get,

$$\tilde{r}_l^n = O(\|u^n\|_{4,\infty} h^2), \quad (h \rightarrow 0). \quad (2-2-5)$$

Let $e_l^n = u(x_l, t^n) - u_l^n$, $\bar{e}_l^{n-1} = u(\bar{x}_l, t^{n-1}) - \bar{u}_l^{n-1}$, then according to (2-1-9) and (2-1-8), we have

$$\begin{cases} \frac{1}{\Delta t}(e_l^n - \bar{e}_l^{n-1}) - \delta_{\bar{x}}(a\delta_x e^n)_l = R_l^n, & 1 \leq l \leq p-1, 1 \leq n \leq m, \\ e_l^0 = 0, & 1 \leq l \leq p-1, \\ e_0^n = 0, e_p^n = 0, & 0 \leq n \leq m, \end{cases} \quad (2-2-6)$$

where $R_l^n = \tilde{r}_l^n - r_l^n$.

By expanding the formula (2-2-6) and organizing it, we can get

$$\left(1 + \frac{2a\Delta t}{h^2}\right) e_l^n = \frac{a\Delta t}{h^2} (e_{l+1}^n + e_{l-1}^n) + R_l^n \Delta t + \bar{e}_l^{n-1}, \quad (2-2-7)$$

where $l = 1, 2, \dots, p-1, n = 1, 2, \dots, m$. Then for any fixed $n (n = 1, 2, \dots, m)$, we have

$$\left(1 + \frac{2a\Delta t}{h^2}\right) |e_l^n| \leq \frac{a\Delta t}{h^2} (\max_l |e_l^n| + \max_l |e_l^n|) + \max_l |R_l^n| \Delta t + \max_l |\bar{e}_l^{n-1}|. \quad (2-2-8)$$

For any given n , by the arbitrariness of l , we can get

$$\left(1 + \frac{2a\Delta t}{h^2}\right) \max_l |e_l^n| \leq \frac{a\Delta t}{h^2} (\max_l |e_l^n| + \max_l |e_l^n|) + \max_l |R_l^n| \Delta t + \max_l |\bar{e}_l^{n-1}|. \quad (2-2-9)$$

After simplification, we have

$$\max_l |e_l^n| \leq \max_l |\bar{e}_l^{n-1}| + \max_l |R_l^n| \Delta t. \quad (2-2-10)$$

Let $I_1\{u_{l-j}^{n-1}, u_{l-j-1}^{n-1}\}$ mean an interpolation of $u(x)$ at the node $(\bar{x}_l, t^{n-1})$ by using u_{l-j}^{n-1} and u_{l-j-1}^{n-1} , similarly we have $I_1\{u(x_{l-j}, t^{n-1}), u(x_{l-j}, t^{n-1})\}$ and $I_1\{e_{l-j}^{n-1}, e_{l-j-1}^{n-1}\}$.

Their specific interpolation coefficients can all be obtained by (2-1-10). Therefore, for $\bar{e}_l^{n-1}$, we have

$$\begin{aligned} \bar{e}_l^{n-1} &= u(\bar{x}_l, t^{n-1}) - \bar{u}_l^{n-1} = u(\bar{x}_l, t^{n-1}) - I_1\{u_{l-j}^{n-1}, u_{l-j-1}^{n-1}\} \\ &= u(\bar{x}_l, t^{n-1}) - I_1\{u_{l-j}^{n-1}, u_{l-j-1}^{n-1}\} - I_1\{u(x_{l-j}, t^{n-1}), u(x_{l-j}, t^{n-1})\} \\ &\quad + I_1\{u(x_{l-j}, t^{n-1}), u(x_{l-j}, t^{n-1})\} \\ &= I_1\{e_{l-j}^{n-1}, e_{l-j-1}^{n-1}\} + u(\bar{x}_l, t^{n-1}) - I_1\{u(x_{l-j}, t^{n-1}), u(x_{l-j}, t^{n-1})\}. \end{aligned} \quad (2-2-11)$$

For any fixed n , according to (2-2-11), we can get

$$\max_l |\bar{e}_l^{n-1}| \leq \max_l |e_l^{n-1}| + \max_l |u(\bar{x}_l, t^{n-1}) - I_1\{u(x_{l-j}, t^{n-1}), u(x_{l-j}, t^{n-1})\}|. \quad (2-2-12)$$

Set

$$\begin{aligned} m(l) &= \{m: |\bar{x}_l - x_m| = \min(|\bar{x}_l - x_{l-j}|, |\bar{x}_l - x_{l-j-1}|)\}, \\ h_l^* &= |\bar{x}_l - x_m| \leq \min\left(\frac{h}{2}, \Delta t b\right). \end{aligned}$$

According to literature [17], the truncation error of the first-order Lagrangian interpolation satisfies

$$|u(\bar{x}_l, t^{n-1}) - I_1\{u(x_{l-j}, t^{n-1}), u(x_{l-j}, t^{n-1})\}| \leq \frac{hh_l^*}{2} \|u^{n-1}\|_{2,\infty}. \quad (2-2-13)$$

Then, by taking (2-2-12) and (2-2-13) back into (2-2-10), we can get

$$\max_l |e_l^n| \leq \max_l |e_l^{n-1}| + \Delta t \left[\frac{1}{2} \|u^{n-1}\|_{2,\infty} \min\left(\frac{h^2}{2\Delta t}, hb\right) + \max_l |R_l^n| \right]. \quad (2-2-14)$$

Let $Y_n = \frac{1}{2} \|u^{n-1}\|_{2,\infty} \min\left(\frac{h^2}{2\Delta t}, hb\right) + \max_l |R_l^n|$, ($n = 1, 2, \dots, m$). According to $e_l^0 = 0$ ($l = 0, 1, \dots, p$), then the following recurrence expression holds

$$\begin{aligned} \max_l |e_l^n| &\leq \max_l |e_l^{n-1}| + \Delta t Y_n \leq \max_l |e_l^{n-2}| + \Delta t (Y_n + Y_{n-1}) \\ &\leq \dots \leq \Delta t \sum_{n=1}^m Y_n \leq \Delta t \frac{T}{\Delta t} \max_n Y_n = T \max_n Y_n. \end{aligned} \quad (2-2-15)$$

Let $\sup_{[0,L] \times [0,T]} \left| \frac{\partial^2 u}{\partial \tau^2} \right| = \left\| \frac{\partial^2 u}{\partial \tau^2} \right\|_\infty$, and take Y_n back into (2-2-15), we can have

$$\begin{aligned} \max_l |e_l^n| &\leq T \max_n \left[\frac{1}{2} \|u^{n-1}\|_{2,\infty} \min\left(\frac{h^2}{2\Delta t}, hb\right) + \max_l |R_l^n| \right] \\ &\leq T \max_n \left[\frac{bh}{2} \|u^{n-1}\|_{2,\infty} + K_1 h^2 \|u^n\|_{4,\infty} + \frac{1+b^2}{2} \Delta t \left\| \frac{\partial^2 u}{\partial \tau^2} \right\|_\infty \right], \end{aligned} \quad (2-2-16)$$

where K_1 is a positive constant. Because n is any given integer from 1 to m , we can get

$$\max_{l,n} |e_l^n| \leq \frac{bh}{2} T \|u\|_{2,\infty} + K_1 T h^2 \|u\|_{4,\infty} + \frac{1+b^2}{2} T \Delta t \left\| \frac{\partial^2 u}{\partial \tau^2} \right\|_\infty. \quad (2-2-17)$$

In summary, we obtain the error analysis result of the one-dimensional linear characteristic difference scheme, and the overall error is the same order as $O(h + \Delta t)$.

2.3 Two-dimensional Linear Characteristic Difference Scheme

In the article [1], the author pointed out that the one-dimensional characteristic difference scheme can be further extended to high dimensions, but the author did not give the specific difference scheme of the high-dimensional characteristic difference scheme in his article. This section will further extend the one-dimensional characteristic difference scheme to two-dimensional scheme.

Firstly, let us consider the following two-dimensional linear convection-dominant diffusion problem

$$\begin{cases} \frac{\partial u}{\partial t} + b_1(x,y) \frac{\partial u}{\partial x} + b_2(x,y) \frac{\partial u}{\partial y} - \nabla \cdot (a(x,y) \nabla u) = f(x,y,t), & (x,y) \in (0,L) \times (0,L), t \in (0,T], \\ u(0,y,t) = g_0(y,t), u(L,y,t) = g_1(y,t), & y \in (0,L), t \in (0,T], \\ u(x,0,t) = g_2(x,t), u(x,L,t) = g_3(x,t), & x \in [0,L], t \in (0,T], \\ u(x,y,0) = u_0(x,y), & (x,y) \in [0,L] \times [0,L], \end{cases} \quad (2-3-1)$$

where L, T are both fixed positive constant, $a(x,y) \geq a_0 > 0$, $\min_{x,y} |b_1(x,y)| \gg$

$\max_{x,y}|a(x,y)|, \min_{x,y}|b_2(x,y)| \gg \max_{x,y}|a(x,y)|$, $u(x,y,t)$ is a smooth function in the domain,

$$\text{and } \nabla \cdot (a(x,y)\nabla u) = \frac{\partial}{\partial x} \left(a(x,y) \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(a(x,y) \frac{\partial u}{\partial y} \right).$$

Let

$$\varphi(x,y) = \sqrt{1 + b_1(x,y)^2 + b_2(x,y)^2}, \quad (2-3-2)$$

Then the characteristic direction accompanying the operator $u_t + b_1(x,y)u_x + b_2(x,y)u_y$ is

$$\tau = \tau(x,y) = \left(\frac{b_1(x,y)}{\varphi(x,y)}, \frac{b_2(x,y)}{\varphi(x,y)}, \frac{1}{\varphi(x,y)} \right). \quad (2-3-3)$$

Taking the directional derivative along the characteristic direction $\tau(x,y)$, then the following operator can be obtained

$$\frac{\partial}{\partial \tau(x,y)} = \frac{b_1(x,y)}{\varphi(x,y)} \frac{\partial}{\partial x} + \frac{b_2(x,y)}{\varphi(x,y)} \frac{\partial}{\partial y} + \frac{1}{\varphi(x,y)} \frac{\partial}{\partial t}. \quad (2-3-4)$$

Bringing (2-3-4) to (2-3-1), we can get the following characteristic equation equivalent to (2-3-1)

$$\begin{cases} \varphi(x,y) \frac{\partial u}{\partial \tau(x,y)} - \nabla \cdot (a(x,y)\nabla u) = f(x,y,t), & (x,y) \in (0,L) \times (0,L), t \in (0,T], \\ u(0,y,t) = g_0(y,t), u(L,y,t) = g_1(y,t), & y \in (0,L), t \in (0,T], \\ u(x,0,t) = g_2(x,t), u(x,L,t) = g_3(x,t), & x \in [0,L], t \in (0,T], \\ u(x,y,0) = u_0(x,y), & (x,y) \in [0,1] \times [0,1]. \end{cases} \quad (2-3-5)$$

Next, we will divide the space evenly, specifically: $x \in [0,L], y \in [0,L]$ are uniformly divided into p parts, and $h = L/p$; $t \in [0,T]$ is divided into m parts uniformly, and $\Delta t = T/m$. (x_l, y_i, t^n) is used to denote the nodes on the grid, where $0 \leq l \leq p, 0 \leq i \leq p, 0 \leq n \leq m$ and l, i, n are all integers.

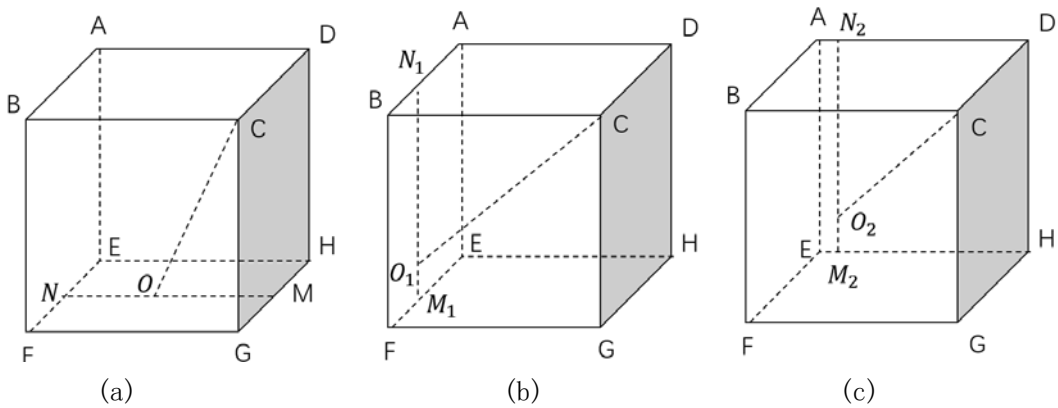


Figure 2-3

Let us assume that $b_1(x,y) > 0, b_2(x,y) > 0$, the rest cases can be derived in a similar way. Consider the point C , which is also node (x_l, y_i, t^n) . At this case, there are three cases of intersection between the characteristic line starting from C and the grid plane (Figure 2-3).

Let $j_1 = j_{1,l,i} = \left\lceil \frac{\Delta t b_1(x_l, y_i)}{h} \right\rceil, j_2 = j_{2,l,i} = \left\lceil \frac{\Delta t b_2(x_l, y_i)}{h} \right\rceil$, then when $l > j_1, i > j_2$, set this case as Case 1. At this case the characteristic line starting from node (x_l, y_i, t^n) intersects the next layer of grid at point O , set it as $(\bar{x}_l, \bar{y}_i, t^{n-1})$ (Figure 2-3(a)).

When $l > j_1, i \leq j_2$ or $l \leq j_1, i \leq j_2$, and $\frac{l}{b_1(x_l, y_i)} \geq \frac{i}{b_2(x_l, y_i)}$, set this case as Case 2, at this case the characteristic line starting from node (x_l, y_i, t^n) intersects the grid plane at point O_1 , set it as (x_{l^*}, y_0, t^{n^*}) (Figure 2-3(b)).

When $l \leq j_1, i > j_2$ or $l \leq j_1, i \leq j_2$, and $\frac{l}{b_1(x_l, y_i)} < \frac{i}{b_2(x_l, y_i)}$, set this case as Case 3, at this case the characteristic line starting from node (x_l, y_i, t^n) intersects the grid plane at point O_2 , set it as (x_0, y_{i^*}, t^{n^*}) (Figure 2-3(c)).

Below we will discuss the three cases separately.

Case 1

When the intersection case of the characteristic line and the grid is Case 1, it is easy to get that the characteristic line will fall in the grid composed of four points $(x_{l-j_1}, y_{i-j_2}, t^{n-1}), (x_{l-j_1-1}, y_{i-j_2}, t^{n-1}), (x_{l-j_1}, y_{i-j_2-1}, t^{n-1}), (x_{l-j_1-1}, y_{i-j_2-1}, t^{n-1})$. Let us set these four points as G, H, F, E . Figure 3-3(a) shows the situation where j_1, j_2 are both zero.

We have known that $|CG| = \Delta t$ and the slope of the characteristic line is $\left(\frac{b_1(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{b_2(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{1}{\varphi(x_l, y_i)} \right)$, then following some simple calculations we can get

$$|GM| = b_1(x_l, y_i) \Delta t, |OM| = b_2(x_l, y_i) \Delta t. \quad (2-3-6)$$

We first give the approximation of $\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}$ at node (x_l, y_i, t^n)

$$\left[\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)} \right]_{x=x_l, y=y_i}^{t=t^n} \approx \varphi(x_l, y_i) \left(\frac{U_C - U_O}{|CO|} \right). \quad (2-3-7)$$

The approximation of the U_O can be obtained by a linear interpolation of U_M and U_N ,

$$U_O \approx \left(\frac{b_2(x_l, y_i) \Delta t}{h} - j_2 \right) U_N + \left(1 - \frac{b_2(x_l, y_i) \Delta t}{h} + j_2 \right) U_M. \quad (2-3-8)$$

The approximation of U_M can be expressed as

$$U_M \approx \left(\frac{b_1(x_l, y_i) \Delta t}{h} - j_1 \right) U_H + \left(1 - \frac{b_1(x_l, y_i) \Delta t}{h} + j_1 \right) U_G, \quad (2-3-9)$$

Similarly, U_N can be expressed as

$$U_N \approx \left(\frac{b_1(x_l, y_i) \Delta t}{h} - j_1 \right) U_E + \left(1 - \frac{b_1(x_l, y_i) \Delta t}{h} + j_1 \right) U_F. \quad (2-3-10)$$

Combining the above three equations, we can get

$$\begin{aligned} U_O &\approx \left(\frac{b_2(x_l, y_i) \Delta t}{h} - j_2 \right) U_N + \left(1 - \frac{b_2(x_l, y_i) \Delta t}{h} + j_2 \right) U_M \\ &\approx \left(\frac{b_2(x_l, y_i) \Delta t}{h} - j_2 \right) \left(\frac{b_1(x_l, y_i) \Delta t}{h} - j_1 \right) U_E + \left(\frac{b_2(x_l, y_i) \Delta t}{h} - j_2 \right) \left(1 - \frac{b_1(x_l, y_i) \Delta t}{h} + j_1 \right) U_F \\ &\quad + \left(1 - \frac{b_2(x_l, y_i) \Delta t}{h} + j_2 \right) \left(\frac{b_1(x_l, y_i) \Delta t}{h} - j_1 \right) U_H + \left(1 - \frac{b_2(x_l, y_i) \Delta t}{h} + j_2 \right) \left(1 - \frac{b_1(x_l, y_i) \Delta t}{h} + j_1 \right) U_G. \end{aligned} \quad (2-3-11)$$

According to $|CG| = \Delta t$ and (2-3-6), we can get $|CO|$ is

$$|CO| = \sqrt{(\Delta t)^2 + |GM|^2 + |OM|^2} = \Delta t \varphi(x_l, y_i). \quad (2-3-12)$$

Combining (2-3-7), (2-3-11), (2-3-12), then we can get the approximation of $\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}$

at node (x_l, y_i, t^n)

$$\begin{aligned} \left[\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)} \right]_{x=x_l, y=y_i}^{t=t^n} &\approx \frac{1}{\Delta t} (U_C - U_O) \\ &\approx \frac{1}{\Delta t} u(x_l, y_i, t^n) - \frac{1}{\Delta t} \left(\frac{b_2(x_l, y_i) \Delta t}{h} - j_2 \right) \left(\frac{b_1(x_l, y_i) \Delta t}{h} - j_1 \right) u(x_{l-j_1-1}, y_{i-j_2-1}, t^{n-1}) \\ &\quad - \frac{1}{\Delta t} \left(\frac{b_2(x_l, y_i) \Delta t}{h} - j_2 \right) \left(1 - \frac{b_1(x_l, y_i) \Delta t}{h} + j_1 \right) u(x_{l-j_1}, y_{i-j_2-1}, t^{n-1}) \\ &\quad - \frac{1}{\Delta t} \left(1 - \frac{b_2(x_l, y_i) \Delta t}{h} + j_2 \right) \left(\frac{b_1(x_l, y_i) \Delta t}{h} - j_1 \right) u(x_{l-j_1-1}, y_{i-j_2}, t^{n-1}) \\ &\quad - \frac{1}{\Delta t} \left(1 - \frac{b_2(x_l, y_i) \Delta t}{h} + j_2 \right) \left(1 - \frac{b_1(x_l, y_i) \Delta t}{h} + j_1 \right) u(x_{l-j_1}, y_{i-j_2}, t^{n-1}). \end{aligned} \quad (2-3-13)$$

The approximation of $\nabla \cdot (a(x, y) \nabla u)_{x=x_l, y=y_i}^{t=t^n}$ can be given by the method of central difference quotient, specifically as

$$\begin{aligned} \nabla \cdot (a(x, y) \nabla u)_{x=x_l, y=y_i}^{t=t^n} &= \left[\frac{\partial}{\partial x} \left(a(x, y) \frac{\partial u}{\partial x} \right) \right]_{x=x_l, y=y_i}^{t=t^n} + \left[\frac{\partial}{\partial y} \left(a(x, y) \frac{\partial u}{\partial y} \right) \right]_{x=x_l, y=y_i}^{t=t^n} \\ &\approx \delta_{\bar{x}} (a(x, y_i) \delta_x u(x, y_i, t^n))_l + \delta_{\bar{y}} (a(x_l, y) \delta_y u(x_l, y, t^n))_i \\ &= \frac{1}{h^2} \left[a \left(x_{l+\frac{1}{2}}, y_i \right) (u(x_{l+1}, y_i, t^n) - u(x_l, y_i, t^n)) - a \left(x_{l-\frac{1}{2}}, y_i \right) (u(x_l, y_i, t^n) - u(x_{l-1}, y_i, t^n)) \right] \end{aligned}$$

$$+ \frac{1}{h^2} \left[a \left(x_l, y_{i+\frac{1}{2}} \right) (u(x_l, y_{i+1}, t^n) - u(x_l, y_i, t^n)) - a \left(x_l, y_{i-\frac{1}{2}} \right) (u(x_l, y_i, t^n) - u(x_l, y_{i-1}, t^n)) \right]. \quad (2-3-14)$$

For equation (2-3-13) and equation (2-3-14), taking the approximate equal sign as an equal sign, using $u, u_{l,i}^n, f_{l,i}^n$ replace $U, u(x_l, y_i, t^n), f(x_l, y_i, t^n)$ respectively and inserting those equations into (2-3-5), we get the difference scheme for Case 1 is

$$\begin{cases} \frac{1}{\Delta t} (u_c - u_o) - \delta_x(a(x, y_i) \delta_x u_i^n)_l - \delta_y(a(x_l, y) \delta_y u_l^n)_i = f_{l,i}^n, \\ \quad \quad \quad 1 \leq l, i \leq p-1, 1 \leq n \leq m, \\ u_{0,i}^n = g_0(y_i, t^n), u_{p,i}^n = g_1(y_i, t^n), \quad 1 \leq i \leq p-1, 1 \leq n \leq m, \\ u_{l,0}^n = g_2(x_l, t^n), u_{l,p}^n = g_3(x_l, t^n), \quad 0 \leq l \leq p, 1 \leq n \leq m, \\ u_{l,i}^0 = u_0(x_l, y_i), \quad 0 \leq l, i \leq p. \end{cases} \quad (2-3-15)$$

Case 2

When the intersection case of the characteristic line and the grid is Case 2, set $j_3 = j_{3l,i} = \left\lceil \frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} \right\rceil$, then we can get that the characteristic line will fall in the grid composed of four points $(x_{l-j_3}, y_0, t^{n-1}), (x_{l-j_3-1}, y_0, t^{n-1}), (x_{l-j_3}, y_0, t^n), (x_{l-j_3-1}, y_0, t^n)$. Let us set these four points as F, E, B, A . Figure 3-3(b) shows the situation where $j_3 = 0, i = 1$.

We have known that $|FG| = ih$ and the slope of the characteristic line is $\left(\frac{b_1(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{b_2(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{1}{\varphi(x_l, y_i)} \right)$, then we can get

$$|N_1 O_1| = \frac{ih}{b_2(x_l, y_i)}, |FM_1| = \frac{ihb_1(x_l, y_i)}{b_2(x_l, y_i)}. \quad (2-3-16)$$

Similarly, we first give the approximation of $\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}$ at node (x_l, y_i, t^n)

$$\left[\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)} \right]_{x=x_l, y=y_i}^{t=t^n} \approx \varphi(x_l, y_i) \left(\frac{U_c - U_{O_1}}{|CO_1|} \right). \quad (2-3-17)$$

The approximation of the U_{O_1} can be obtained by a linear interpolation of U_{M_1} and U_{N_1} ,

$$U_{O_1} \approx \left(1 - \frac{ih}{b_2(x_l, y_i) \Delta t} \right) U_{N_1} + \frac{ih}{b_2(x_l, y_i) \Delta t} U_{M_1}. \quad (2-3-18)$$

The approximation of U_{M_1} can be expressed as

$$U_{M_1} \approx \left(\frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} - j_3 \right) U_E + \left(1 - \frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} + j_3 \right) U_F, \quad (2-3-19)$$

Similarly, for U_{N_1} we have

$$U_{N_1} \approx \left(\frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} - j_3 \right) U_A + \left(1 - \frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} + j_3 \right) U_B. \quad (2-3-20)$$

Combining above three equations, we can get

$$\begin{aligned}
 U_{O_1} &\approx \left(1 - \frac{ih}{b_2(x_l, y_i)\Delta t}\right) U_{N_1} + \frac{ih}{b_2(x_l, y_i)\Delta t} U_{M_1} \\
 &\approx \left(1 - \frac{ih}{b_2(x_l, y_i)\Delta t}\right) \left(\frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} - j_3\right) U_A + \left(1 - \frac{ih}{b_2(x_l, y_i)\Delta t}\right) \left(1 - \frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} + j_3\right) U_B \\
 &\quad + \frac{ih}{b_2(x_l, y_i)\Delta t} \left(\frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} - j_3\right) U_E + \frac{ih}{b_2(x_l, y_i)\Delta t} \left(1 - \frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} + j_3\right) U_F.
 \end{aligned} \tag{2-3-21}$$

According to $|FG| = ih$ and (2-3-16), then $|CO_1|$ equals

$$|CO_1| = \sqrt{(ih)^2 + |N_1O_1|^2 + |FM_1|^2} = \frac{ih\varphi(x_l, y_i)}{b_2(x_l, y_i)}. \tag{2-3-22}$$

Combining (2-3-17), (2-3-21), (2-3-22), the approximation of $\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}$ at the node (x_l, y_i, t^n) can be expressed as

$$\begin{aligned}
 \left[\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}\right]_{x=x_l, y=y_i}^{t=t^n} &\approx \frac{b_2(x_l, y_i)}{ih} (U_C - U_{O_1}) \\
 &\approx \frac{b_2(x_l, y_i)}{ih} u(x_l, y_i, t^n) - \frac{b_2(x_l, y_i)}{ih} \left(1 - \frac{ih}{b_2(x_l, y_i)\Delta t}\right) \left(\frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} - j_3\right) u(x_{l-j_3-1}, y_0, t^n) \\
 &\quad - \frac{b_2(x_l, y_i)}{ih} \left(1 - \frac{ih}{b_2(x_l, y_i)\Delta t}\right) \left(1 - \frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} + j_3\right) u(x_{l-j_3}, y_0, t^n) \\
 &\quad - \frac{1}{\Delta t} \left(\frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} - j_3\right) u(x_{l-j_3-1}, y_0, t^{n-1}) - \frac{1}{\Delta t} \left(1 - \frac{ib_1(x_l, y_i)}{b_2(x_l, y_i)} + j_3\right) u(x_{l-j_3}, y_0, t^{n-1}).
 \end{aligned} \tag{2-3-23}$$

The approximation method and result of $\nabla \cdot (a(x, y)\nabla u)_{x=x_l, y=y_i}^{t=t^n}$ are the same as in Case 1, shown as (2-3-14).

For equation (2-3-23) and equation (2-3-14), taking the approximate equal sign as an equal sign, using $u, u_{l,i}^n, f_{l,i}^n$ replace $U, u(x_l, y_i, t^n), f(x_l, y_i, t^n)$ respectively and inserting those equations into the (2-3-5), we get the difference scheme for Case 2 is

$$\begin{cases} \frac{b_2(x_l, y_i)}{ih} (u_C - u_{O_1}) - \delta_x(a(x, y_i)\delta_x u_i^n)_l - \delta_y(a(x_l, y)\delta_y u_l^n)_i = f_{l,i}^n, \\ \quad \quad \quad 1 \leq l, i \leq p-1, 1 \leq n \leq m, \\ u_{0,i}^n = g_0(y_i, t^n), u_{p,i}^n = g_1(y_i, t^n), \quad 1 \leq i \leq p-1, 1 \leq n \leq m, \\ u_{l,0}^n = g_2(x_l, t^n), u_{l,p}^n = g_3(x_l, t^n), \quad 0 \leq l \leq p, 1 \leq n \leq m, \\ u_{l,i}^0 = u_0(x_l, y_i), \quad 0 \leq l, i \leq p. \end{cases} \tag{2-3-24}$$

Case 3

When the intersection case of the characteristic line and the grid is Case 3, set $j_4 = j_{4,l,i} = \left\lfloor \frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} \right\rfloor$, then we can get that the characteristic line will fall in the grid composed of four points $(x_0, y_{i-j_4}, t^{n-1}), (x_0, y_{i-j_4-1}, t^{n-1}), (x_0, y_{i-j_4}, t^n), (x_0, y_{i-j_4-1}, t^n)$. Let us set these four points as H, E, D, A . Figure 3-3(c) shows the situation where $j_4 = 0, l = 1$.

We have known that $|GH| = lh$ and the slope of the characteristic line is $\left(\frac{b_1(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{b_2(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{1}{\varphi(x_l, y_i)}\right)$, then we can get

$$|N_2 O_2| = \frac{lh}{b_1(x_l, y_i)}, |M_2 H| = \frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)}. \quad (2-3-25)$$

Similarly as in Case 1 and Case 2, we first give the approximation of $\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}$ at node (x_l, y_i, t^n)

$$\left[\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}\right]_{x=x_l, y=y_i}^{t=t^n} \approx \varphi(x_l, y_i) \left(\frac{U_C - U_{O_2}}{|CO_2|}\right). \quad (2-3-26)$$

The approximation of the U_{O_2} can be obtained by a linear interpolation of U_{M_2} and U_{N_2} ,

$$U_{O_2} \approx \left(1 - \frac{lh}{b_1(x_l, y_i)\Delta t}\right) U_{N_2} + \frac{lh}{b_1(x_l, y_i)\Delta t} U_{M_2}. \quad (2-3-27)$$

The approximation of U_{M_2} can be expressed as

$$U_{M_2} \approx \left(\frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} - j_4\right) U_E + \left(1 - \frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} + j_4\right) U_H, \quad (2-3-28)$$

Similarly, for U_{N_2} we have

$$U_{N_2} \approx \left(\frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} - j_4\right) U_A + \left(1 - \frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} + j_4\right) U_D. \quad (2-3-29)$$

Combining above three equations, we can get

$$\begin{aligned} U_{O_2} &\approx \left(1 - \frac{lh}{b_1(x_l, y_i)\Delta t}\right) U_{N_2} + \frac{lh}{b_1(x_l, y_i)\Delta t} U_{M_2} \\ &\approx \left(1 - \frac{lh}{b_1(x_l, y_i)\Delta t}\right) \left(\frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} - j_4\right) U_A + \left(1 - \frac{lh}{b_1(x_l, y_i)\Delta t}\right) \left(1 - \frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} + j_4\right) U_D \\ &\quad + \frac{lh}{b_1(x_l, y_i)\Delta t} \left(\frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} - j_4\right) U_E + \frac{lh}{b_1(x_l, y_i)\Delta t} \left(1 - \frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} + j_4\right) U_H. \end{aligned} \quad (2-3-30)$$

According to $|GH| = lh$ and (2-3-16), then $|CO_2|$ equals

$$|CO_2| = \sqrt{(lh)^2 + |N_2 O_2|^2 + |M_2 H|^2} = \frac{lh\varphi(x_l, y_i)}{b_1(x_l, y_i)}. \quad (2-3-31)$$

Combining (2-3-26), (2-3-30), (2-3-31), the approximation of $\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}$ at the node (x_l, y_i, t^n) can be expressed as

$$\begin{aligned} \left[\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}\right]_{x=x_l, y=y_i}^{t=t^n} &\approx \frac{b_1(x_l, y_i)}{lh} (U_C - U_{O_2}) \\ &\approx \frac{b_1(x_l, y_i)}{lh} u(x_l, y_i, t^n) - \frac{b_1(x_l, y_i)}{lh} \left(1 - \frac{lh}{b_1(x_l, y_i)\Delta t}\right) \left(\frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} - j_4\right) u(x_0, y_{i-j_4-1}, t^n) \\ &\quad - \frac{b_1(x_l, y_i)}{lh} \left(1 - \frac{lh}{b_1(x_l, y_i)\Delta t}\right) \left(1 - \frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} + j_4\right) u(x_0, y_{i-j_4}, t^n) \end{aligned}$$

$$-\frac{1}{\Delta t} \left(\frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} - j_4 \right) u(x_0, y_{i-j_4-1}, t^{n-1}) - \frac{1}{\Delta t} \left(1 - \frac{lb_2(x_l, y_i)}{b_1(x_l, y_i)} + j_4 \right) u(x_0, y_{i-j_4}, t^{n-1}). \quad (2-3-32)$$

The approximation method and result of $\nabla \cdot (a(x, y) \nabla u)_{x=x_l, y=y_i}^{t=t^n}$ are the same as in Case 1 and Case 2, shown as (2-3-14).

For equation (2-3-32) and equation (2-3-14), taking the approximate equal sign as an equal sign, using $u, u_{l,i}^n, f_{l,i}^n$ replace $U, u(x_l, y_i, t^n), f(x_l, y_i, t^n)$ respectively and inserting those equations into the (2-3-5), we get the difference scheme for Case 3 is

$$\begin{cases} \frac{b_1(x_l, y_i)}{lh} (u_c - u_{o_2}) - \delta_x(a(x, y_i) \delta_x u_l^n)_l - \delta_y(a(x_l, y) \delta_y u_l^n)_i = f_{l,i}^n, \\ \quad \quad \quad 1 \leq l, i \leq p-1, 1 \leq n \leq m, \\ u_{0,i}^n = g_0(y_i, t^n), u_{p,i}^n = g_1(y_i, t^n), \quad 1 \leq i \leq p-1, 1 \leq n \leq m, \\ u_{l,0}^n = g_2(x_l, t^n), u_{l,p}^n = g_3(x_l, t^n), \quad 0 \leq l \leq p, 1 \leq n \leq m, \\ u_{l,i}^0 = u_0(x_l, y_i), \quad 0 \leq l, i \leq p. \end{cases} \quad (2-3-33)$$

We have discussed the three cases separately. Next we will unify the above three cases into one expression. First, let $s_1 = s_{1,l,i} = \text{sign}(l - j_1)$, $s_2 = s_{2,l,i} = \text{sign}(i - j_2)$, $s_3 = s_{3,l,i} = \text{sign}(l/b_1(x_l, y_i) - i/b_2(x_l, y_i))$. Then according to the conditions that appear in the three cases, let

$$J_1 = J_1(x_l, y_i) = s_1 \frac{(1+s_1)}{2} s_2 \frac{(1+s_2)}{2}. \quad (2-3-34)$$

Then if and only when $l > j_1, i > j_2$, $J_1 = 1$, if not, then $J_1 = 0$. When $J_1 = 1$, the intersection case of the characteristic line and the grad is Case 1. Let

$$\begin{aligned} J_2 = J_2(x_l, y_i) &= s_1 \frac{(1+s_1)}{2} \left(1 - s_2 \frac{(1+s_2)}{2} \right) \\ &+ \left(1 - s_1 \frac{(1+s_1)}{2} \right) \left(1 - s_2 \frac{(1+s_2)}{2} \right) \left(1 - s_3 \frac{(s_3-1)}{2} \right). \end{aligned} \quad (2-3-35)$$

Then if and only when $l > j_1, i \leq j_2$ or $l \leq j_1, i \leq j_2$, and $\frac{l}{b_1(x_l, y_i)} \geq \frac{i}{b_2(x_l, y_i)}$, $J_2 = 1$, if not, then $J_2 = 0$. When $J_2 = 1$, the intersection case of the characteristic line and the grad is Case 2. Let

$$\begin{aligned} J_3 = J_3(x_l, y_i) &= \left(1 - s_1 \frac{(1+s_1)}{2} \right) s_2 \frac{(1+s_2)}{2} \\ &+ \left(1 - s_1 \frac{(1+s_1)}{2} \right) \left(1 - s_2 \frac{(1+s_2)}{2} \right) \left(s_3 \frac{(s_3-1)}{2} \right). \end{aligned} \quad (2-3-36)$$

Then if and only when $l \leq j_1, i > j_2$ or $l \leq j_1, i \leq j_2$, and $\frac{l}{b_1(x_l, y_i)} < \frac{i}{b_2(x_l, y_i)}$, $J_3 = 1$, if

Overall, we can unify the two-dimensional linear characterisitic difference scheme into one expression

$$\begin{cases} \frac{J_1(u_C - u_{O_0})}{\Delta t} + \frac{2J_2 b_2 (u_C - u_{O_1})}{ih} + \frac{2J_3 b_1 (u_C - u_{O_2})}{lh} - \delta_x(a(x, y_i) \delta_x u_i^n)_l - \delta_y(a(x_l, y) \delta_y u_l^n)_i = f_{ln}^n, \\ \quad \quad \quad 1 \leq l, i \leq p-1, 1 \leq n \leq m, \\ u_{0,i}^n = g_0(y_i, t^n), u_{p,i}^n = g_1(y_i, t^n), \quad 1 \leq i \leq p-1, 1 \leq n \leq m, \\ u_{l,0}^n = g_2(x_l, t^n), u_{l,p}^n = g_3(x_l, t^n), \quad 0 \leq l \leq p, 1 \leq n \leq m, \\ u_{l,i}^0 = u_0(x_l, y_i), \quad 0 \leq l, i \leq p. \end{cases} \quad (2-3-37)$$

Chapter 3 One-dimensional Improved Characteristic Difference Scheme

3. 1 Comments on Characteristic Difference Scheme

In the previous chapter, we introduced the characteristic difference scheme in detail and did the error analysis of the one-dimensional characteristic difference scheme with constant coefficients. In this section, we will point out some problems that the characteristic difference scheme brings. We shall still consider the following one-dimensional linear convection-dominated convection-diffusion equation

$$\begin{cases} \frac{\partial u}{\partial t} + b(x) \frac{\partial u}{\partial x} - \frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right) = f(x, t), & (x, t) \in (0, L) \times (0, T], \\ u(x, 0) = u_0(x), & x \in (0, L), \\ u(0, t) = g_0(t), u(L, t) = g_1(t), & t \in (0, T]. \end{cases} \quad (3-1-1)$$

where $a(x) \geq a_0 > 0$, $\min_x |b(x)| \gg \max_x a(x)$, and $u(x, t)$ is smooth in its domain. As discussed in Chapter 2, by introducing the characteristic difference technique, an equation equivalent to the above equation can be obtained

$$\begin{cases} \varphi(x) \frac{\partial u}{\partial \tau(x)} - \frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right) = f(x, t), & (x, t) \in (0, L) \times (0, T], \\ u(x, 0) = u_0(x), & x \in (0, L), \\ u(0, t) = g_0(t), u(L, t) = g_1(t), & t \in (0, T]. \end{cases} \quad (3-1-2)$$

The main advantage of the above characteristic difference scheme is reflected in the time truncation error. In section 2.2, we have proved that the time truncation error of the characteristic difference scheme under constant coefficients is $\frac{1+b^2}{2} T \Delta t \left\| \frac{\partial^2 u}{\partial \tau^2} \right\|_{\infty}$, while usually the time truncation error of the difference scheme of equation (3-1-1) is $A \Delta t \left\| \frac{\partial^2 u}{\partial t^2} \right\|_{\infty}$, where A is a positive constant. In paper [1], the author pointed out that when the equation is convectively dominant, the change in the solution of the equation along the characteristic direction τ is smaller than the change along the time direction t . Therefore, the characteristic difference scheme is often more advantageous than other methods in solving the convection-dominated convection diffusion equation. However, the time truncation error of the characteristic difference scheme is also proportional to $1 + b^2$, so

when the convection term coefficient b increases, the error generated by using the characteristic difference scheme will also increase accordingly. In the following, we have verified this through a set of numerical experiments, and the results show that when the convection term coefficient b increases, the calculation effect of the characteristic difference scheme deteriorated drastically, resulting in serious numerical diffusion.

There is an intuitive explanation for this phenomenon as follows: When the convection term coefficient b is large, the slope of the characteristic line is small, and the intersection of the characteristic line and the next layer may be far away from the node from which the characteristic line starts (Figure 3-1). At this case, the length of the characteristic line intercepted by the upper and lower layers of the grid is longer, so the approximate effect of $\varphi(x) \frac{\partial u}{\partial \tau(x)}$ will be worse, which makes the characteristic difference scheme appear more serious numerical diffusion.

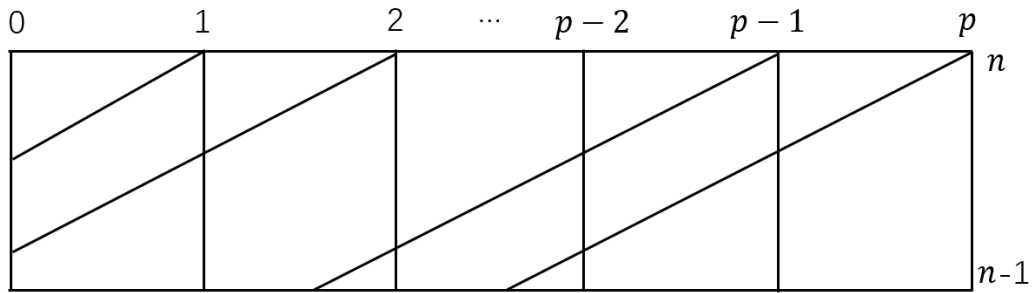


Figure 3-1

When the ratio of the time step Δt to the space step h is small, this problem is not important, because the error in the space direction will be dominant under this case. But an important application of the characteristic difference scheme is to solve large time steps question. Therefore, from the above analysis, the b value has a great influence on the calculation accuracy of the characteristic difference scheme. Therefore, it is necessary to carry out special processing on the characteristic difference scheme when b is large.

Since in the characteristic difference scheme, the important reason for this problem is that the intersection of the characteristic line and the next layer is far away from the node

from which the characteristic line starts, resulting in a poor approximation of $\varphi(x) \frac{\partial u}{\partial \tau(x)}$, so a natural idea is to shorten the distance between the intersection of the characteristic line and the grid and the node where the characteristic line starts.

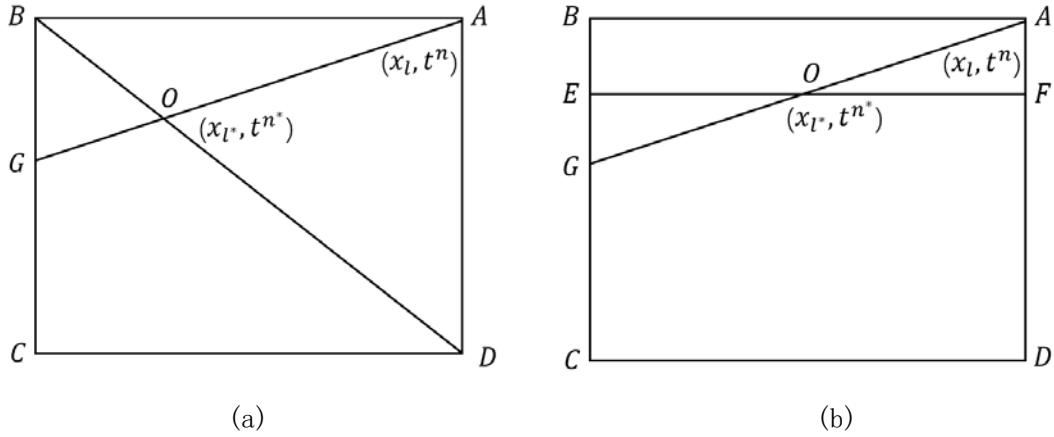


Figure 3-2

A possible method is described as new characteristic difference scheme in [2]. For all nodes (x_l, t^n) , the intersection point O between the diagonal of the first grid passed by the characteristic line and the characteristic line is used to replace the intersection point between the characteristic line and the next layer. Meanwhile the u value at this O point and the u value at the node (x_l, t^n) is used to approximate $\left(\varphi(x) \frac{\partial u}{\partial \tau(x)}\right)_l^n$. The u value at point O is obtained by linearly interpolating by using the value of u at two points B, D (Figure 3-2(a)). Subsequent numerical experiment results show that although this scheme avoids the problems caused by the characteristic difference scheme to a certain extent, its numerical diffusion is still large. This may be due to the fact that when $\left(\varphi(x) \frac{\partial u}{\partial \tau(x)}\right)_l^n$ is approximated, the error contained in the approximate value of u on the time layer t^n is greater than the error contained in the approximate value of u on the time layer t^{n-1} . At the same time, this scheme ignores the superiority of the characteristic difference scheme when the coefficient of the convection term is small. Thus, in this case the fitting effect of the new characteristic difference scheme is worse than that of the characteristic difference

scheme.

This paper will construct another difference scheme called the improved characteristic difference scheme (Figure 3-2(b)). When the convection term coefficient at the node (x_l, t^n) is large, specifically: when $b(x_l) > h/\Delta t$. We consider the case when the characteristic line starting from node (x_l, t^n) intersects the grid at point G , and take the midpoint E of BG to make a straight line EF which is parallel to CD . Then, we can use the value of u at point A and O to approximate $\left(\varphi(x) \frac{\partial u}{\partial \tau(x)}\right)_l^n$, where the value of u at point O is obtained by the linear interpolation at the two points E and F , while the values at the two points E and F are given by B, C and A, D respectively. In the following, we will give detailed expressions to explain the above approximation method. When the convection term coefficient at the node (x_l, t^n) is small, that is, when $b(x_l) \leq h/\Delta t$. This is the case when the characteristic line does not intersect the left grid at point G , and the effect of the characteristic difference scheme is better than that of selecting other points for approximation. Therefore, in the above case, we still consider the approximation method to $\left(\varphi(x) \frac{\partial u}{\partial \tau(x)}\right)_l^n$ in characteristic difference scheme.

The advantage of using the improved characteristic difference scheme is that it solves the large numerical diffusion problem of the characteristic difference scheme when the convection term coefficient is large, and it also combines the advantage of the characteristic difference scheme with small time truncation error when the convection coefficient is small. In the following text, we prove that when we use the improved characteristic difference scheme, the time truncation error of this scheme is $\frac{\Delta t}{2} T \left\| \frac{\partial^2 u}{\partial t^2} \right\|_\infty$, which has nothing to do with the convection term coefficient b . Therefore, although $\left\| \frac{\partial^2 u}{\partial t^2} \right\|_\infty$ is larger than $\left\| \frac{\partial^2 u}{\partial \tau^2} \right\|_\infty$, $\frac{\Delta t}{2} T \left\| \frac{\partial^2 u}{\partial t^2} \right\|_\infty$ is still smaller compared to $\frac{1+b^2}{2} T \Delta t \left\| \frac{\partial^2 u}{\partial \tau^2} \right\|_\infty$ with large b .

Subsequent numerical experiments show that, compared with the new characteristic difference scheme in [2], the improved characteristic difference scheme is much better than the new characteristic difference scheme when solving equations with small convection term coefficients. At the same time, when solving equations with large convection term

coefficients, the improved characteristic difference scheme still has smaller calculation errors than the new characteristic difference scheme.

3.2 One-dimensional Improved Characteristic Difference Scheme

3.2.1 Linear Case

Next, we give the detailed derivation process of the one-dimensional linear improved characteristic difference scheme. We only consider the case of $b(x_l) > h/\Delta t$, because the case of $b(x_l) \leq h/\Delta t$ has been discussed in detail in the Chapter 2.

Consider the equation (3-1-2), when the characteristic line and the grid intersect at the point G (Figure 3-2(b)). We firstly give the approximation of $\varphi(x) \frac{\partial u}{\partial \tau(x)}$ at the node (x_l, t^n) . Let $|BG| = \Delta t'$, according to $|EF| = h$ and the slope of the characteristic line is $\frac{1}{b(x_l)}$. It is easy to get

$$\Delta t' = \frac{h}{b(x_l)}. \quad (3-2-1)$$

Let us use $I_1\{\cdot, \cdot\}$ to represent a linear interpolation. Then we have

$$\begin{aligned} U_E &\approx I_1\{u(x_{l-1}, t^n), u(x_{l-1}, t^{n-1})\} \\ &= \frac{\Delta t'}{2\Delta t} u(x_{l-1}, t^{n-1}) + \left(1 - \frac{\Delta t'}{2\Delta t}\right) u(x_{l-1}, t^n). \end{aligned} \quad (3-2-2)$$

Similarly,

$$\begin{aligned} U_F &\approx I_1\{u(x_l, t^n), u(x_l, t^{n-1})\} \\ &= \frac{\Delta t'}{2\Delta t} u(x_l, t^{n-1}) + \left(1 - \frac{\Delta t'}{2\Delta t}\right) u(x_l, t^n). \end{aligned} \quad (3-2-3)$$

Using $I_2\{\cdot, \cdot\}$ to represent a linear interpolation different from $I_1\{\cdot, \cdot\}$ interpolation, then we can get

$$\begin{aligned} U_O &\approx I_2\{U_E, U_F\} = \frac{1}{2}U_E + \frac{1}{2}U_F \\ &\approx \frac{h}{4b(x_l)\Delta t} u(x_{l-1}, t^{n-1}) + \left(\frac{1}{2} - \frac{h}{4b(x_l)\Delta t}\right) u(x_{l-1}, t^n) \\ &\quad + \frac{h}{4b(x_l)\Delta t} u(x_l, t^{n-1}) + \left(\frac{1}{2} - \frac{h}{4b(x_l)\Delta t}\right) u(x_l, t^n). \end{aligned} \quad (3-2-4)$$

For $|AO|$, we can get

$$|AO| = \sqrt{\frac{h^2}{4} + \frac{\Delta t'^2}{4}} = \frac{h\varphi}{2b(x_l)}. \quad (3-2-5)$$

So, for the above case of $b(x_l) > h/\Delta t$, the approximation of $\left(\varphi(x) \frac{\partial u}{\partial \tau(x)}\right)_l^n$ can be

u_l^{n-1} can be regarded as a known value, so the processing of the nonlinear equation (3-2-8) is completely similar to the processing of the linear equation (3-1-1). Let $s = s_{l,n} = \text{sign}(\Delta tb(x_l, u_l^{n-1})/c(x_l, u_l^{n-1}) - h)$, then we can get the non-linear improved characteristic difference scheme

$$\begin{cases} \frac{s(s+1)b(x_l, u_l^{n-1})}{h}(u_l^n - u_l^{n*}) + \left(1 - \frac{s(s+1)}{2}\right) \frac{c(x_l, u_l^{n-1})}{\Delta t}(u_l^n - \bar{u}_l^{n-1}) - \delta_{\bar{x}}(a(x)\delta_x u^n)_l = f_l^n, \\ u_l^0 = u_0(x_l), \quad 1 \leq l \leq p-1, \\ u_0^n = g_1(t^n), \quad u_p^n = g_2(t^n), \quad 0 \leq n \leq m, \end{cases} \quad 1 \leq l \leq p-1, 1 \leq n \leq m, \quad (3-2-9)$$

where

$$\begin{aligned} u_l^{n*} = & \frac{hc(x_l, u_l^{n-1})}{4b(x_l, u_l^{n-1})\Delta t} u_{l-1}^{n-1} + \left(\frac{1}{2} - \frac{hc(x_l, u_l^{n-1})}{4b(x_l, u_l^{n-1})\Delta t}\right) u_{l-1}^n \\ & + \frac{hc(x_l, u_l^{n-1})}{4b(x_l, u_l^{n-1})\Delta t} u_l^{n-1} + \left(\frac{1}{2} - \frac{hc(x_l, u_l^{n-1})}{4b(x_l, u_l^{n-1})\Delta t}\right) u_l^n, \end{aligned} \quad (3-2-10)$$

$$\bar{u}_l^{n-1} = \left(\frac{\Delta tb(x_l, u_l^{n-1})}{c(x_l, u_l^{n-1})h}\right) u_{l-1}^{n-1} + \left(1 - \frac{\Delta tb(x_l, u_l^{n-1})}{c(x_l, u_l^{n-1})h}\right) u_l^{n-1}. \quad (3-2-11)$$

3. 3 Error Analysis

Next, we will perform error analysis on the improved characteristic difference scheme with constant coefficients. Assuming that $b > 0$, since when $b < 0$, the proof method is similar to that when $b > 0$. Note that when $\Delta tb \leq h$, the formula (3-2-7) is the characteristic difference scheme

$$\begin{cases} \frac{1}{\Delta t}(u_l^n - \bar{u}_l^{n-1}) - \delta_{\bar{x}}(a\delta_x u^n)_l = f_l^n, \quad 1 \leq l \leq p-1, 1 \leq n \leq m, \\ u_l^0 = u_0(x_l), \quad 1 \leq l \leq p-1, \\ u_0^n = g_1(t^n), \quad u_l^n = g_2(t^n), \quad 0 \leq n \leq m. \end{cases} \quad (3-3-1)$$

The error analysis process in this case has been given in detail in section 2.2.

Next, we only consider the case of when $\Delta tb > h$. From (3-2-7), when $s = 1$, we have the following equation

$$\begin{cases} \frac{2b}{h}(u_l^n - u_l^{n*}) - \delta_{\bar{x}}(a\delta_x u^n)_l = f_l^n, \quad 1 \leq l \leq p-1, 1 \leq n \leq m, \\ u_l^0 = u_0(x_l), \quad 1 \leq l \leq p-1, \\ u_0^n = g_1(t^n), \quad u_l^n = g_2(t^n), \quad 0 \leq n \leq m. \end{cases} \quad (3-3-2)$$

Set

$$r_l^n = \varphi \left(\frac{\partial u}{\partial \tau} \right)_l^n - \frac{2b}{h} (u(x_l, t^n) - u(x_{l^*}, t^{n*})). \quad (3-3-3)$$

Using $(x(\tau), t(\tau))$ to represent the coordinates of the point on the characteristic line, and taking the Taylor expansion of $u(x_{l^*}, t^{n*})$ along the τ direction in node (x_l, t^n) , where

the remainder takes the form of integral remainder, we can get

$$u(x_{l^*}, t^{n^*}) = u(x_l, t^n) - \frac{h\varphi}{2b} \left(\frac{\partial u}{\partial \tau} \right)_l^n - \int_{(x_l, t^n)}^{(x_{l^*}, t^{n^*})} \sqrt{(x(\tau) - x_{l^*})^2 + (t(\tau) - t^{n^*})^2} \frac{\partial^2 u}{\partial \tau^2} d\tau. \quad (3-3-4)$$

By several calculations, we have

$$r_l^n = \frac{(1+b^2)}{4b} \left(\frac{\partial^2 u}{\partial \tau^2} \right)_{l^*}^{n^*} h, \quad (3-3-5)$$

where $\left(\frac{\partial^2 u}{\partial \tau^2} \right)_{l^*}^{n^*}$ is the value of the second-order tangent derivative along the line AB at a certain point on AB.

Set

$$\tilde{r}_l^n = \left[\frac{\partial}{\partial x} \left(a \frac{\partial u}{\partial x} \right) \right]_l^n - \delta_{\bar{x}}(a \delta_x u(x, t^n))_l. \quad (3-3-6)$$

Taking the Taylor expansion of $u(x_{l+1}, t^n), u(x_{l-1}, t^n)$ along the x direction at the node (x_l, t^n) separately, we can get

$$u(x_{l+1}, t^n) = u(x_l, t^n) + h \left(\frac{\partial u}{\partial x} \right)_l^n + \frac{h^2}{2} \left(\frac{\partial^2 u}{\partial x^2} \right)_l^n + \frac{h^3}{3!} \left(\frac{\partial^3 u}{\partial x^3} \right)_l^n + \frac{h^4}{4!} \left(\frac{\partial^4 u}{\partial x^4} \right)_l^n + o(h^4), \quad (3-3-7)$$

$$u(x_{l-1}, t^n) = u(x_l, t^n) - h \left(\frac{\partial u}{\partial x} \right)_l^n + \frac{h^2}{2} \left(\frac{\partial^2 u}{\partial x^2} \right)_l^n - \frac{h^3}{3!} \left(\frac{\partial^3 u}{\partial x^3} \right)_l^n + \frac{h^4}{4!} \left(\frac{\partial^4 u}{\partial x^4} \right)_l^n + o(h^4). \quad (3-3-8)$$

Bringing formula (3-3-7) and formula (3-3-8) back into formula (3-3-6), so we have

$$\begin{aligned} \tilde{r}_l^n &= \left[\frac{\partial}{\partial x} \left(a \frac{\partial u}{\partial x} \right) \right]_l^n - \delta_{\bar{x}}(a \delta_x u(x, t^n))_l \\ &= a \left(\frac{\partial^2 u}{\partial x^2} \right)_l^n - \frac{a}{h^2} (u(x_{l-1}, t^n) + u(x_{l+1}, t^n) - 2u(x_l, t^n)) \\ &= -\frac{ah^2}{12} \left(\frac{\partial^4 u}{\partial x^4} \right)_l^n + o(h^2) = O(\|u^n\|_{4,\infty} h^2). \end{aligned} \quad (3-3-9)$$

Putting formula (3-3-3) and formula (3-3-6) back into formula (3-1-2), we can get

$$\begin{cases} \frac{2b}{h} (u(x_l, t^n) - u(x_{l^*}, t^{n^*})) - \delta_{\bar{x}}(a \delta_x u(x, t^n))_l = f(x_l, t^n) + R_l^n, & 1 \leq l \leq p-1, 1 \leq n \leq m, \\ u(x_l, 0) = u_0(x_l), & 1 \leq l \leq p-1, \\ u(0, t^n) = g_1(t^n), u(L, t^n) = g_2(t^n), & 0 \leq n \leq m, \end{cases} \quad (3-3-10)$$

where $R_l^n = \tilde{r}_l^n - r_l^n$. Setting $e_l^n = u(x_l, t^n) - u_l^n$, $e_{l^*}^{n^*} = u(x_{l^*}, t^{n^*}) - u_{l^*}^{n^*}$, and considering (3-3-2), (3-3-10), we can get

$$\begin{cases} \frac{2b}{h} (e_l^n - e_{l^*}^{n^*}) - \delta_{\bar{x}}(a \delta_x e^n)_l = R_l^n, & 1 \leq l \leq p-1, 1 \leq n \leq m, \\ e_l^0 = 0, & 1 \leq l \leq p-1, \\ e_0^n = 0, e_p^n = 0, & 0 \leq n \leq m. \end{cases} \quad (3-3-11)$$

Expand (3-3-11), then

$$\left(2b + \frac{2a}{h} \right) e_l^n = \frac{a}{h} (e_{l+1}^n + e_{l-1}^n) + R_l^n h + 2b e_{l^*}^{n^*}, \quad (3-3-12)$$

where $l = 1, 2, \dots, p-1, n = 1, 2, \dots, m$. For any fixed n , we can have the following inequality

$$\left(2b + \frac{2a}{h}\right) |e_l^n| \leq \frac{a}{h} \left(\max_l |e_l^n| + \max_l |e_l^n|\right) + \max_l |R_l^n| h + 2b \max_l |e_l^{n*}|. \quad (3-3-13)$$

For any fixed $n (n = 1, 2, \dots, m)$, according to the arbitrariness of l , we have

$$\left(2b + \frac{2a}{h}\right) \max_l |e_l^n| \leq \frac{a}{h} \left(\max_l |e_l^n| + \max_l |e_l^n|\right) + \max_l |R_l^n| h + 2b \max_l |e_l^{n*}|. \quad (3-3-14)$$

After simplifying the above formula, we can get

$$\max_l |e_l^n| \leq \max_l |e_l^{n*}| + \frac{1}{2b} \max_l |R_l^n| h. \quad (3-3-15)$$

Next, the following further processing is performed on $\max_l |e_l^{n*}|$ to construct an inequality relationship between $\max_l |e_l^n|$ and $\max_l |e_l^{n-1}|$. For e_l^{n*} , the following equation holds

$$\begin{aligned} e_l^{n*} &= u(x_l^*, t^{n*}) - u_l^{n*} = u(x_l^*, t^{n*}) - I_2\{u_{l-1}^*, u_l^{n*}\} \\ &= u(x_l^*, t^{n*}) - I_2\{I_1\{u_{l-1}^n, u_{l-1}^{n-1}\}, I_1\{u_l^n, u_l^{n-1}\}\} \\ &= u(x_l^*, t^{n*}) - I_2\{I_1\{u_{l-1}^n, u_{l-1}^{n-1}\}, I_1\{u_l^n, u_l^{n-1}\}\} \\ &\quad + I_2\{I_1\{u(x_{l-1}, t^n), u(x_{l-1}, t^{n-1})\}, I_1\{u(x_l, t^n), u(x_l, t^{n-1})\}\} \\ &\quad - I_2\{I_1\{u(x_{l-1}, t^n), u(x_{l-1}, t^{n-1})\}, I_1\{u(x_l, t^n), u(x_l, t^{n-1})\}\} \\ &= u(x_l^*, t^{n*}) - I_2\{I_1\{u(x_{l-1}, t^n), u(x_{l-1}, t^{n-1})\}, I_1\{u(x_l, t^n), u(x_l, t^{n-1})\}\} \\ &\quad + I_2\{I_1\{e_{l-1}^n, e_{l-1}^{n-1}\}, I_1\{e_l^n, e_l^{n-1}\}\}, \end{aligned} \quad (3-3-16)$$

where $I_1\{\cdot, \cdot\}$ and $I_2\{\cdot, \cdot\}$ represent one-dimensional linear interpolation with different coefficients. The specific interpolation scheme can be seen in (3-2-2) and (3-2-4).

Setting $\tilde{R}_l^n = u(x_l^*, t^{n*}) - I_2\{I_1\{u(x_{l-1}, t^n), u(x_{l-1}, t^{n-1})\}, I_1\{u(x_l, t^n), u(x_l, t^{n-1})\}\}$, then according to (3-3-16) and $\Delta t b > h$, for fixed n , we can get

$$\max_l |e_l^{n*}| \leq \max_l |\tilde{R}_l^n| + \frac{h}{2b\Delta t} \max_l |e_l^{n-1}| + \left(1 - \frac{h}{2b\Delta t}\right) \max_l |e_l^n|. \quad (3-3-17)$$

Taking (3-3-15) into the above inequality, after simplifying, we can get

$$\max_l |e_l^{n*}| \leq \max_l |e_l^{n-1}| + \frac{2b\Delta t}{h} \max_l |\tilde{R}_l^n| + \left(\Delta t - \frac{h}{2b}\right) \max_l |R_l^n|. \quad (3-3-18)$$

In formula (3-3-18), $\tilde{R}_l^n$ is the interpolation error. For the truncation error generated by using Lagrange interpolation, there is a very detailed introduction in the literature [17], we will use it directly. Let

$$m(n) = \{m: |t^{n*} - t^m| = \min(|t^{n*} - t^n|, |t^{n*} - t^{n-1}|)\},$$

$$\rho^n = |t^{n*} - t^m| \leq \min\left(\frac{\Delta t}{2}, \frac{h}{2b}\right),$$

and set $\left\|\frac{\partial^2 u}{\partial t^2}\right\|_\infty = \max_{[0,L] \times [0,T]} \left|\frac{\partial^2 u}{\partial t^2}\right|$. Thus, for the truncation error generated by using Lagrange interpolation, we have

$$|u(x_{l-1}, t^{n*}) - I_1\{u(x_{l-1}, t^n), u(x_{l-1}, t^{n-1})\}| \leq \frac{\Delta t \rho^n}{2} \left\|\frac{\partial^2 u}{\partial t^2}\right\|_\infty. \quad (3-3-19)$$

Similarly,

$$|u(x_l, t^{n*}) - I_1\{u(x_l, t^n), u(x_l, t^{n-1})\}| \leq \frac{\Delta t \rho^n}{2} \left\|\frac{\partial^2 u}{\partial t^2}\right\|_\infty, \quad (3-3-20)$$

and

$$|u(x_{l^*}, t^{n*}) - I_2\{u(x_l, t^{n*}), u(x_{l-1}, t^{n*})\}| \leq \frac{1}{2!} \|u\|_{2,\infty} \cdot \frac{h}{2} \cdot \frac{h}{2} = \frac{h^2}{8} \|u\|_{2,\infty}. \quad (3-3-21)$$

Then we can get

$$\begin{aligned} |\tilde{R}_l^n| &= |u(x_{l^*}, t^{n*}) - I_2\{I_1\{u(x_{l-1}, t^n), u(x_{l-1}, t^{n-1})\}, I_1\{u(x_l, t^n), u(x_l, t^{n-1})\}\})| \\ &= |u(x_{l^*}, t^{n*}) - I_2\{u(x_l, t^{n*}), u(x_{l-1}, t^{n*})\} + I_2\{u(x_l, t^{n*}), u(x_{l-1}, t^{n*})\} \\ &\quad - I_2\{I_1\{u(x_{l-1}, t^n), u(x_{l-1}, t^{n-1})\}, I_1\{u(x_l, t^n), u(x_l, t^{n-1})\}\})| \\ &\leq |u(x_{l^*}, t^{n*}) - I_2\{u(x_l, t^{n*}), u(x_{l-1}, t^{n*})\}| \\ &\quad + \frac{1}{2} |u(x_{l-1}, t^{n*}) - I_1\{u(x_{l-1}, t^n), u(x_{l-1}, t^{n-1})\}| \\ &\quad + \frac{1}{2} |u(x_l, t^{n*}) - I_1\{u(x_l, t^n), u(x_l, t^{n-1})\}| \leq \frac{h^2}{8} \|u\|_{2,\infty} + \frac{\Delta t \rho^n}{2} \left\|\frac{\partial^2 u}{\partial t^2}\right\|_\infty, \end{aligned} \quad (3-3-22)$$

where $l = 1, 2, \dots, p-1$, n is a fixed integer between 1 and m . According to the arbitrariness of l , we get

$$\max_l |\tilde{R}_l^n| \leq \frac{h^2}{8} \|u\|_{2,\infty} + \frac{\Delta t \rho^n}{2} \left\|\frac{\partial^2 u}{\partial t^2}\right\|_\infty. \quad (3-3-23)$$

Bringing (3-3-23), (3-3-18) into (3-3-15), the following inequalities hold

$$\begin{aligned} \max_l |e_l^n| &\leq \max_l |e_l^{n-1}| + \frac{2b\Delta t}{h} \max_l |\tilde{R}_l^n| + \Delta t \max_l |R_l^n| \\ &\leq \max_l |e_l^{n-1}| + \frac{bh\Delta t}{4} \|u\|_{2,\infty} + \frac{\Delta t^2}{2} \left\|\frac{\partial^2 u}{\partial t^2}\right\|_\infty + \Delta t \max_l |R_l^n| \\ &\leq \max_l |e_l^{n-1}| + \Delta t \left(\frac{bh}{4} \|u\|_{2,\infty} + \frac{\Delta t}{2} \left\|\frac{\partial^2 u}{\partial t^2}\right\|_\infty + \max_l |R_l^n| \right). \end{aligned} \quad (3-3-24)$$

In the second inequality of the above inequalities, the fact that ρ^n is less than or equal to $\frac{h}{2b}$ is used. Let $Y_n = \frac{bh}{4} \|u\|_{2,\infty} + \frac{\Delta t}{2} \left\|\frac{\partial^2 u}{\partial t^2}\right\|_\infty + \max_l |R_l^n|$ ($n = 1, 2, \dots, m$), then according to $e_l^0 = 0$ ($l = 0, 1, \dots, p$) and (3-3-24), the following recurrence expression

holds

$$\begin{aligned} \max_l |e_l^n| &\leq \max_l |e_l^{n-1}| + \Delta t Y_n \leq \max_l |e_l^{n-2}| + \Delta t (Y_n + Y_{n-1}) \\ &\leq \dots \leq \Delta t \sum_{n=1}^m Y_n \leq \Delta t \frac{T}{\Delta t} \max_n Y_n = T \max_n Y_n. \end{aligned} \quad (3-3-25)$$

Set $\left\| \frac{\partial^2 u}{\partial \tau^2} \right\|_\infty = \max_{[0,L] \times [0,T]} \left| \frac{\partial^2 u}{\partial \tau^2} \right|$ and bring Y_n into (3-3-25), we have

$$\begin{aligned} \max_l |e_l^n| &\leq T \max_n \left(\frac{bh}{4} \|u\|_{2,\infty} + \frac{\Delta t}{2} \left\| \frac{\partial^2 u}{\partial t^2} \right\|_\infty + \max_l |R_l^n| \right) \\ &\leq T \max_n \left[\left(\frac{b}{4} \|u\|_{2,\infty} + \frac{(1+b^2)}{4b} \left\| \frac{\partial^2 u}{\partial \tau^2} \right\|_\infty \right) h + K_2 h^2 \|u\|_{4,\infty} + \frac{\Delta t}{2} \left\| \frac{\partial^2 u}{\partial t^2} \right\|_\infty \right], \end{aligned} \quad (3-3-26)$$

where K_2 is a positive constant. Since n is any given positive integer from 1 to m , it can be further sorted to get the final result

$$\max_{l,n} |e_l^n| \leq \left(\frac{b}{4} \|u\|_{2,\infty} + \frac{(1+b^2)}{4b} \left\| \frac{\partial^2 u}{\partial \tau^2} \right\|_\infty \right) Th + K_2 Th^2 \|u\|_{4,\infty} + \frac{\Delta t}{2} T \left\| \frac{\partial^2 u}{\partial t^2} \right\|_\infty. \quad (3-3-27)$$

Overall, we get the error estimation result of the improved characteristic difference scheme. The overall error is the same order as $O(h + \Delta t)$, which is consistent with the error estimation result of the characteristic difference scheme. At the same time, the time truncation of the scheme is independent of the convection term coefficient b .

3. 4 Numerical Experiments

In this paper, a total of four numerical experiments are carried out on the improved one-dimensional characteristic difference scheme. The convergence of the linear improved characteristic difference scheme, the advantage of the improved characteristic difference scheme compared with the characteristic difference scheme and the new characteristic difference scheme, and the convergence and superiority of nonlinear improved characteristic difference scheme are verified respectively.

3. 4. 1 Experiment 1

We shall use the following example [18] to verify the convergence of the improved one-dimensional characteristic difference scheme.

$$\begin{cases} \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \varepsilon \frac{\partial^2 u}{\partial x^2}, & -\infty < x < \infty, t > 0, \\ u(x,0) = \exp\left(-\frac{(x-x_0)^2}{2\rho_0^2}\right), & -\infty < x < \infty, \end{cases} \quad (3-4-1)$$

where a, ε are both constant, and $\varepsilon > 0$. The analytical solution of this problem is

$$u(x,t) = \frac{\rho_0}{\rho} \exp \left[-\frac{(x-x_0-at)^2}{2\rho_0^2} \right], \quad (3-4-2)$$

where $\rho^2 = \rho_0^2 + \varepsilon t$. Let us take $\rho_0 = 20, x_0 = 100$. Next, we are going to calculate the numerical solution of the equation when $0 < x < 300, 0 < t < 2$. Set $a = 5$, take $\varepsilon = 0.05$ and $\varepsilon = 0.005$ separately, and we can get Table 3-1.

Table 3-1

ε	h	Δt	$E_\infty(h, \Delta t)$	Order
0.05	1/2	1/4	0.0153	*
	1/4	1/8	0.0078	1.96
	1/8	1/16	0.0040	1.95
	1/16	1/32	0.0021	1.90
0.005	1/2	1/4	0.0152	*
	1/4	1/8	0.0077	1.97
	1/8	1/16	0.0039	1.97
	1/16	1/32	0.0020	1.95

where

$$E_\infty(h, \Delta t) = \max_{\substack{0 \leq l \leq p \\ 0 \leq n \leq m}} |u(x_l, t^n) - u_l^n|, \text{ Order} = E_\infty(2h, 2\Delta t) / E_\infty(h, \Delta t).$$

From the above table, it can be seen that when the time step and space step are reduced to 1/2 of the original, the error is also reduced to about half of the original. That is, the one-dimensional linear improved characteristic difference scheme we constructed is first-order convergence in time and space. This result is consistent with our proof in Section 3.3. Further, the improved characteristic difference scheme has a very good convergence even with small diffusion coefficient (i. e. $a \gg \varepsilon$).

3.4.2 Experiment 2

Consider the following system of equations [19]

$$\begin{cases} \frac{\partial u}{\partial t} + b \frac{\partial u}{\partial x} - a \frac{\partial^2 u}{\partial x^2} = f(x, t), & 0 < x < 2, 0 < t \leq 5, \\ u(x, 0) = u_0(x), & 0 < x < 2, \\ u(0, t) = g_1(t), u(2, t) = g_2(t), & 0 \leq t \leq 5, \end{cases} \quad (3-4-3)$$

where a, b are constants and $a > 0$. Suppose the analytical solution of this equation is $u(x, t) = e^x \sin(x - t)$, then the right term $f(x, t)$ and initial boundary value

$u_0(x), g_1(t), g_2(t)$ can be given by the analytical solution and (3-4-3).

Consider the calculation effects of the characteristic difference scheme, the new characteristic difference scheme and the improved characteristic difference scheme of the (3-4-3) when $h = 0.05, \Delta t = 0.05, a = 0.001$ and $b = 1, b = 5, b = 10$ separately.

The following Figure 3-3 (a), (b), (c) are the numerical solution and analytical solution curves of the characteristic difference scheme when $a = 0.001$, and $b = 1, b = 5, b = 10$ respectively. The realization and the dotted line of these figures represent the analytical solution and numerical solution curve respectively at time $t = 5$. It is not difficult to see that when $b = 1$, the characteristic difference scheme is used to obtain good numerical results, and the numerical solution curve is very close to the analytical solution curve. Experimental data shows that when $t = 5$, the maximum error is only 0.0358. But when $b = 5$, the use of the characteristic difference scheme has a serious numerical diffusion phenomenon. Further, when $b = 5$, as shown in Figure 3-3(c), serious numerical diffusion occurs, which makes the numerical calculation results extremely inaccurate. The above results are the same as what we discussed in Section 3.1.

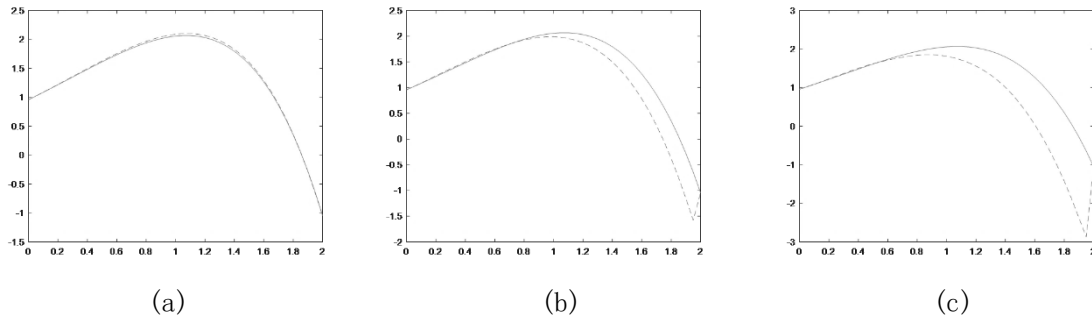


Figure 3-3

The following Figure 3-4 (a), (b), (c) are the numerical solution and analytical solution curves when $a = 0.001$ and $b = 1, b = 5, b = 10$ respectively. The realization and the dotted line of these figures represent the analytical solution and numerical solution curve respectively at time $t = 5$. It can be seen that when $b = 1$, the effect of using this scheme to find the numerical solution of the equation is much worse than the characteristic difference scheme. In fact, the numerical results show that when $t = 5$, the maximum error

produced by using this scheme is 0.2626, which is much larger than the error given by using the characteristic difference scheme. When $b = 5, b = 10$, using this scheme to calculate the equation, the maximum errors obtained at $t = 5$ are 0.2628 and 0.2453 respectively. It can be seen that, compared with the characteristic difference scheme, the use of this scheme has avoided serious numerical diffusion when the convection term coefficient is large. However, the numerical results calculated using this scheme are still not ideal, especially when the convection term coefficient $b = 1$, the numerical results are far worse than the characteristic difference scheme.

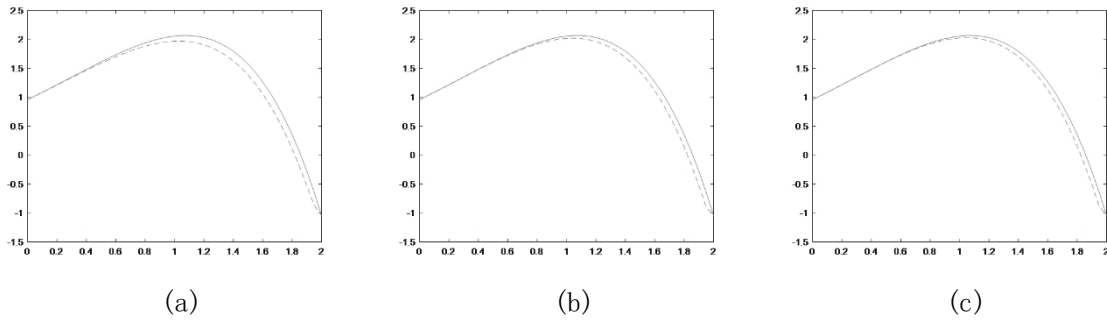


Figure 3-4

The following Figure 3-5 (a), (b), (c) are the numerical solution and analytical solution curves when $a = 0.001$ and $b = 1, b = 5, b = 10$ respectively. The realization and the dotted line of these figures represent the analytical solution and numerical solution curve respectively at time $t = 5$. When $b = 1$, the improved characteristic difference scheme happens to be the characteristic difference scheme, and the numerical error is very small in this case. The numerical solution curve and the analytical solution curve almost overlap. At the same time, when $b = 5$ and $b = 10$, this scheme avoids the serious numerical diffusion phenomenon that occurs when using the characteristic difference scheme. And the numerical results show that when $b = 5, b = 10$, the maximum error obtained at time $t = 5$ is 0.2263 and 0.2269 respectively. The accuracy is higher than that obtained by the new characteristic difference scheme.

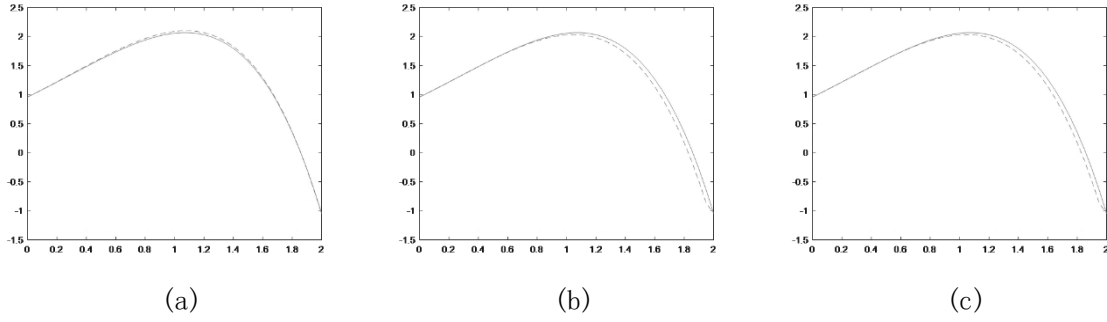


Figure 3-5

The above experimental results can be sorted into Table 3-2.

Table 3-2

a	b	$E^1_{\infty}(h, \Delta t)$	$E^2_{\infty}(h, \Delta t)$	$E^3_{\infty}(h, \Delta t)$
0.001	1	0.0358	0.2626	0.0358
	5	0.9354	0.2628	0.2263
	10	2.2270	0.2453	0.2269

where

$$E^i_{\infty}(h, \Delta t) = \max_{\substack{0 \leq l \leq p \\ n=m}} |u^i(x_l, t^n) - u_l^n|, (i = 1, 2, 3).$$

$i = 1, 2, 3$ represent the characteristic difference scheme, the new characteristic difference scheme and the improved characteristic difference scheme respectively.

From the above figures and tables, we can draw the following conclusions: The characteristic difference scheme has excellent numerical results when the convection term coefficient is small. But when the convection term coefficient increases, the error produced by the characteristic difference scheme increases sharply. The new characteristic difference scheme solves this problem, but its numerical accuracy is not ideal. The improved characteristic difference scheme has excellent numerical results when the convection term coefficient is small. At the same time, when the coefficient of the convection term increases, it avoids serious numerical oscillations, and has smaller calculation errors than the new characteristic difference scheme.

3.4.3 Experiment 3

In order to numerically verify the convergence of the one-dimensional nonlinear improved characteristic difference scheme constructed in Section 3.2.2, we shall consider the nonlinear convection-diffusion equation

$$\begin{cases} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} - a \frac{\partial^2 u}{\partial x^2} = f(x, t), & 0 < x < 1, 0 < t < 1, \\ u(x, 0) = u_0(x), & 0 < x < 1, \\ u(0, t) = g_1(t), u(1, t) = g_2(t), & 0 \leq t \leq 1, \end{cases} \quad (3-4-4)$$

where $0 < a = \text{const} \ll 1$. Suppose the analytical solution of this equation is $u = 1 + \frac{1 - \exp(-\frac{(1-x)t}{a})}{1 - \exp(-\frac{1}{a})}$, then the right term $f(x, t)$ and initial boundary value

$u_0(x), g_1(t), g_2(t)$ can be given as follows

$$\begin{cases} f(x, t) = \frac{\exp(-\frac{(1-x)t}{a})}{a[1 - \exp(-\frac{1}{a})]} \cdot \left[(1-x) - t \left(1 + \frac{1 - \exp(-\frac{(1-x)t}{a})}{1 - \exp(-\frac{1}{a})} \right) + t^2 \right], \\ g_1(t) = 1 + \frac{1 - \exp(-\frac{t}{a})}{1 - \exp(-\frac{1}{a})}, \\ u_0(x) = g_2(t) = 1. \end{cases} \quad (3-4-5)$$

Considering the calculation result of the improved characteristic difference scheme when $a = 0.05, 0.005$, Table 3-3 can be obtained.

Table 3-3

a	$h, \Delta t$	$E_\infty(h, \Delta t)$	Order
0.05	1/200	0.0203	*
	1/400	0.0301	1.71
	1/800	0.0153	1.80
	1/600	0.0077	1.88
0.005	1/200	0.5150	*
	1/400	0.2970	1.73
	1/800	0.1605	1.85
	1/1600	0.0838	1.92

The definitions of $E_\infty(h, \Delta t)$ and Order are the same as those in Table 3-1.

From the above table, when the time and space step size is reduced to 1/2 of the original, the error is also reduced to about half of the original. So we numerically verify that the error

of the nonlinear improved characteristic difference scheme is generally equal to $O(\Delta t + h)$. At the same time, when the convection term coefficient is very small, the improved characteristic difference scheme also avoids numerical oscillations, and still has good convergence.

Furthermore, the calculation effects of the characteristic difference scheme, the new characteristic difference scheme and the improved characteristic difference scheme on the equation are considered respectively. Let $h = 0.01, \Delta t = 0.05$, $a = 0.05, 0.005$ separately, we can get Table 3-4.

Table 3-4

a	$E^1_{\infty}(h, \Delta t)$	$E^2_{\infty}(h, \Delta t)$	$E^3_{\infty}(h, \Delta t)$
0.05	0.2339	0.0731	0.0702
0.005	0.6803	0.2523	0.2452

The definition of $E^i_{\infty}(h, \Delta t)$ is the same as that in Table 3-2.

From Table 3-4, we can see that the calculation effect of the improved f characteristic difference scheme is better than that of the characteristic difference scheme and the new characteristic difference scheme. The characteristic difference scheme has the largest numerical error when calculating the above equation. This is because at the above case, when the feature difference format is used, the length of the characteristic line intercepted by the upper and lower grid lines is longer, resulting in poor approximation of the characteristic derivative. The improved characteristic difference scheme absorbs the advantage of characteristic difference scheme when approximates $\varphi(x) \frac{\partial u}{\partial \tau(x)}$ under small convection coefficient. At the same time, when approximating $\varphi(x) \frac{\partial u}{\partial \tau(x)}$ at the case $u(x_l, t^{n-1}) > 1/5$, the improved characteristic difference scheme does not use the intersection of the characteristic line and the next layer, but choose the point O , thereby reducing the approximate error of the derivative.

3.4.4 Experiment 4

Consider the following nonlinear convection-diffusion equation

$$\begin{cases} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = v \frac{\partial^2 u}{\partial x^2}, & -1 < x < 0, 0 < t < 2, \\ u(x, 0) = g(x), & -1 < x < 0, \\ u(-1, t) = 1, u(0, t) = 0, & 0 \leq t \leq 2, \end{cases} \quad (3-4-6)$$

where $g(x)$ is given by the boundary conditions $u(-1, t) = 1, u(1, t) = 0$ by linear interpolation, that is, $g(x) = -x$, and the static analytical solution of the problem is

$$f(x) = \frac{1 - e^{\frac{x}{v}}}{1 + e^{\frac{x}{v}}}.$$

Consider the numerical results obtained by the improved characteristic difference scheme to process the above equation when the diffusion term coefficient $v = 0.005$. Taking $h = 0.001, \Delta t = 0.005$, Figure 3-6 can be obtained, where the solid line represents the static analytical solution of the problem, The dashed line from left to right represents the numerical solution curves at time $t = i/10, (i = 0, 1, \dots, 20)$. It can be seen from the solid line in the Figure 3-6, the smoothness of the solution curve of this equation is poor. When $x \rightarrow 0$, the function value drops rapidly from 1 to 0. It can be seen from the dotted line in the Figure 3-6 that as time increases, the numerical solution curves gradually approach the static analytical solution curve, and when $t = 2$, the curve of the numerical solution almost coincides with the curve of the analytical solution. This shows that the nonlinear improved characteristic difference scheme can still obtain good numerical results when dealing with equations with poor smoothness.

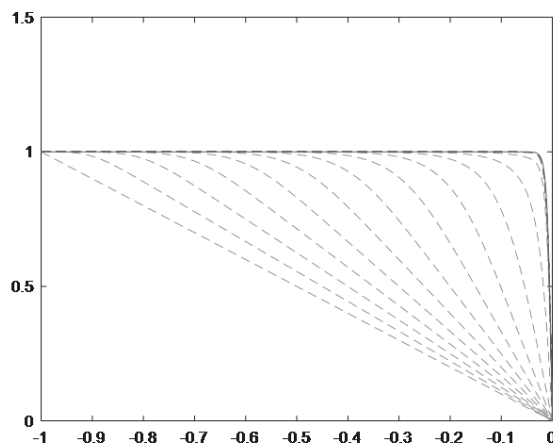


Figure 3-6

Chapter 4 Two-dimensional Linear Improved Characteristic Difference Scheme

4. 1 Two-dimensional Linear Improved Characteristic Difference Scheme

In this chapter, we will further extend the improved characteristic difference scheme to two-dimensional problems. Like the two-dimensional characteristic difference scheme, let us consider the following two-dimensional convection-dominated diffusion problem

$$\begin{cases} \frac{\partial u}{\partial t} + b_1(x,y)\frac{\partial u}{\partial x} + b_2(x,y)\frac{\partial u}{\partial y} - \nabla \cdot (a(x,y)\nabla u) = f(x,y,t), & (x,y) \in (0,L) \times (0,L), t \in (0,T], \\ u(0,y,t) = g_0(y,t), u(L,y,t) = g_1(y,t), & y \in (0,L), t \in (0,T], \\ u(x,0,t) = g_2(x,t), u(x,L,t) = g_3(x,t), & x \in [0,L], t \in (0,T], \\ u(x,y,0) = u_0(x,y), & (x,y) \in [0,L] \times [0,L]. \end{cases} \quad (4-1-1)$$

where L, T are fixed positive number, $a(x,y) \geq a_0 > 0, \min_{x,y} |b_1(x,y)| \gg \max_{x,y} |a(x,y)|, \min_{x,y} |b_2(x,y)| \gg \max_{x,y} |a(x,y)|$ and $u(x,y,t)$ is a smooth function in its domain, $\nabla \cdot (a(x,y)\nabla u) = \frac{\partial}{\partial x} \left(a(x,y) \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(a(x,y) \frac{\partial u}{\partial y} \right)$. As in Section 2.3, by introducing characteristic difference technology, formula (4-1-1) can be transformed into the following equivalent equation

$$\begin{cases} \varphi(x,y) \frac{\partial u}{\partial \tau(x,y)} - \nabla \cdot (a(x,y)\nabla u) = f(x,y,t), & (x,y) \in (0,L) \times (0,L), t \in (0,T], \\ u(0,y,t) = g_0(y,t), u(L,y,t) = g_1(y,t), & y \in (0,L), t \in (0,T], \\ u(x,0,t) = g_2(x,t), u(x,L,t) = g_3(x,t), & x \in [0,L], t \in (0,T], \\ u(x,y,0) = u_0(x,y), & (x,y) \in [0,L] \times [0,L]. \end{cases} \quad (4-1-2)$$

Let us assume that $b_1(x,y) > 0, b_2(x,y) > 0$, and the rest cases can be constructed in a similar way. Then the characteristic line starting from point C may have three cases as shown in Figure 4-1(a), (b), (c). Let these three cases be Case 1, Case 2, and Case 3. Below we shall discuss these three cases separately.

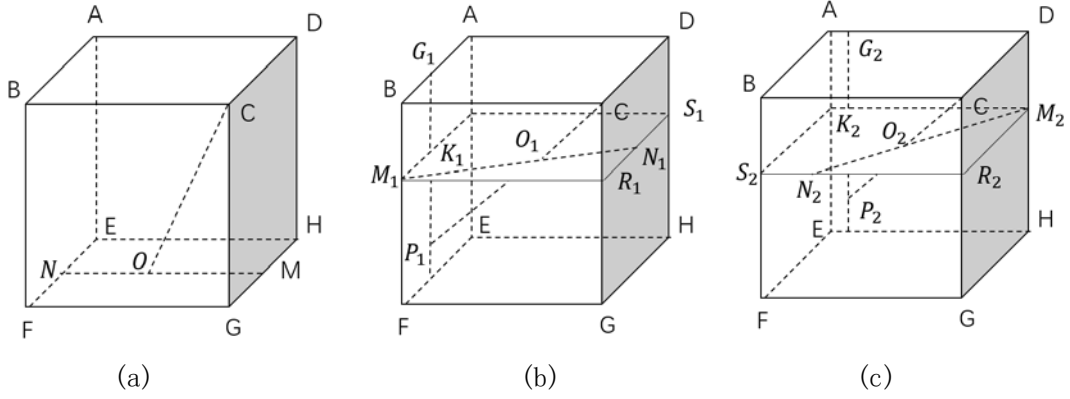


Figure 4-1

Case 1

Case 1 is shown as Figure 4-1(a), at this case $b_1(x_l, y_i), b_2(x_l, y_i)$ satisfy $b_1(x_l, y_i) \leq h/\Delta t$ and $b_2(x_l, y_i) \leq h/\Delta t$. The characteristic line starting from point C intersects the grid plane $EHFG$ at point O . Across point O , we can draw a parallel line of EH and FG . The line intersects EF and HG at points N and M respectively.

We have known that $|CG| = \Delta t$ and the slope of characteristic line is $\left(\frac{b_1(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{b_2(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{1}{\varphi(x_l, y_i)}\right)$, so we have

$$|GM| = b_1(x_l, y_i)\Delta t, |OM| = b_2(x_l, y_i)\Delta t. \quad (4-1-3)$$

We firstly give the approximation of $\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}$ at node (x_l, y_i, t^n)

$$\left[\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}\right]_{x=x_l, y=y_i}^{t=t^n} \approx \varphi(x_l, y_i) \frac{U_C - U_O}{|CO|}. \quad (4-1-4)$$

The approximation of U_O can be obtained by linear interpolation of U_M and U_N ,

$$U_O \approx \frac{b_2(x_l, y_i)\Delta t}{h} U_N + \left(1 - \frac{b_2(x_l, y_i)\Delta t}{h}\right) U_M. \quad (4-1-5)$$

Further, according to (4-1-3), we have

$$U_M \approx \frac{b_1(x_l, y_i)\Delta t}{h} U_H + \left(1 - \frac{b_1(x_l, y_i)\Delta t}{h}\right) U_G, \quad (4-1-6)$$

Similarly,

$$U_N \approx \frac{b_1(x_l, y_i)\Delta t}{h} U_E + \left(1 - \frac{b_1(x_l, y_i)\Delta t}{h}\right) U_F. \quad (4-1-7)$$

Combining the three above equations, we can get

$$U_O \approx \frac{b_2(x_l, y_i)\Delta t}{h} U_N + \left(1 - \frac{b_2(x_l, y_i)\Delta t}{h}\right) U_M$$

$$\begin{aligned} &\approx \frac{b_2(x_l, y_i) b_1(x_l, y_i) \Delta t^2}{h^2} U_E + \frac{b_2(x_l, y_i) \Delta t}{h} \left(1 - \frac{b_1(x_l, y_i) \Delta t}{h}\right) U_F \\ &+ \frac{b_1(x_l, y_i) \Delta t}{h} \left(1 - \frac{b_2(x_l, y_i) \Delta t}{h}\right) U_H + \left(1 - \frac{b_2(x_l, y_i) \Delta t}{h}\right) \left(1 - \frac{b_1(x_l, y_i) \Delta t}{h}\right) U_G. \end{aligned} \quad (4-1-8)$$

According to $|CG| = \Delta t$ and (4-1-3), we have

$$|CO| = \sqrt{(\Delta t)^2 + |GM|^2 + |OM|^2} = \Delta t \varphi(x_l, y_i). \quad (4-1-9)$$

Then, combining (4-1-4), (4-1-8), (4-1-9), the following holds

$$\begin{aligned} \left[\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)} \right]_{x=x_l, y=y_i}^{t=t^n} &\approx \frac{1}{\Delta t} (U_C - U_O) \\ &\approx \frac{1}{\Delta t} u(x_l, y_i, t^n) - \frac{b_2(x_l, y_i) b_1(x_l, y_i) \Delta t}{h^2} u(x_{l-1}, y_{i-1}, t^{n-1}) \\ &\quad - \frac{b_2(x_l, y_i)}{h} \left(1 - \frac{b_1(x_l, y_i) \Delta t}{h}\right) u(x_l, y_{i-1}, t^{n-1}) \\ &\quad - \frac{b_1(x_l, y_i)}{h} \left(1 - \frac{b_2(x_l, y_i) \Delta t}{h}\right) u(x_{l-1}, y_i, t^{n-1}) \\ &\quad - \frac{1}{\Delta t} \left(1 - \frac{b_2(x_l, y_i) \Delta t}{h}\right) \left(1 - \frac{b_1(x_l, y_i) \Delta t}{h}\right) u(x_{l-1}, y_{i-1}, t^{n-1}). \end{aligned} \quad (4-1-10)$$

The approximate method for $\nabla \cdot (a(x, y) \nabla u)_{x=x_l, y=y_i}^{t=t^n}$ is given by the method of central difference quotient, specific deduction process and results are consistent with formula (2-3-14).

For equations (4-1-10) and (2-3-14), taking the approximate equal sign as an equal sign, using $u, u_{l,i}^n, f_{l,i}^n$ replace $U, u(x_l, y_i, t^n), f(x_l, y_i, t^n)$, and bringing them into (4-1-2), then the difference format of the Case 1 can be obtained as

$$\begin{cases} \frac{1}{\Delta t} (u_C - u_O) - \delta_{\bar{x}}(a(x, y_i) \delta_x u_i^n)_l - \delta_{\bar{y}}(a(x_l, y) \delta_y u_l^n)_i = f_{l,i}^n, \\ \quad \quad \quad 1 \leq l, i \leq p-1, 1 \leq n \leq m, \\ u_{0,i}^n = g_0(y_i, t^n), u_{p,i}^n = g_1(y_i, t^n), \quad 1 \leq i \leq p-1, 1 \leq n \leq m, \\ u_{l,0}^n = g_2(x_l, t^n), u_{l,p}^n = g_3(x_l, t^n), \quad 0 \leq l \leq p, 1 \leq n \leq m, \\ u_{l,i}^0 = u_0(x_l, y_i), \quad 0 \leq l, i \leq p. \end{cases} \quad (4-1-11)$$

Case 2

Case 2 is shown in Figure 4-1(b), at this case $b_1(x_l, y_i), b_2(x_l, y_i)$ satisfy $b_2(x_l, y_i) > h/\Delta t$ and $b_1(x_l, y_i) \leq b_2(x_l, y_i)$. The characteristic line starting from point C intersects the grid plane $ABFE$ at point P_1 . Across point P_1 , we can draw a parallel line of AE and BF . The line intersects AB at point G_1 . Taking the midpoint of $P_1 G_1$ as K_1 , acrossing K_1 , we can draw a plane parallel to the $EFGH$ as $M_1 S_1 R_1$. Set the characteristic line and plane $M_1 S_1 R_1$ intersects at point O_1 , then connects point M_1 and point O_1 and extends the line

to intersect S_1R_1 at point N_1 .

We have known that $|FG| = h$ and the slope of characteristic line is $\left(\frac{b_1(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{b_2(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{1}{\varphi(x_l, y_i)}\right)$, so we have

$$|G_1P_1| = \frac{h}{b_2(x_l, y_i)}, |R_1N_1| = \frac{hb_1(x_l, y_i)}{b_2(x_l, y_i)}. \quad (4-1-12)$$

We firstly give the approximation of $\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}$ at node (x_l, y_i, t^n)

$$\left[\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}\right]_{x=x_l, y=y_i}^{t=t^n} \approx \varphi(x_l, y_i) \frac{U_C - U_{O_1}}{|CO_1|}. \quad (4-1-13)$$

From the fact that K_1 is the midpoint of line segment P_1G_1 , then according to geometric knowledge, point O_1 is also the midpoint of line segment M_1N_1 , so obviously

$$U_{O_1} \approx \frac{1}{2}U_{M_1} + \frac{1}{2}U_{N_1}. \quad (4-1-14)$$

The approximation of U_{M_1} can be obtained by linear interpolation of U_B and U_F ,

$$U_{M_1} \approx \frac{h}{2b_2(x_l, y_i)\Delta t}U_F + \left(1 - \frac{h}{2b_2(x_l, y_i)\Delta t}\right)U_B. \quad (4-1-15)$$

Similarly

$$U_{N_1} \approx \frac{b_1(x_l, y_i)}{b_2(x_l, y_i)}U_{S_1} + \left(1 - \frac{b_1(x_l, y_i)}{b_2(x_l, y_i)}\right)U_{R_1}. \quad (4-1-16)$$

The approximation of U_{S_1} can be obtained by linear interpolation of U_D and U_H ,

$$U_{S_1} \approx \frac{h}{2b_2(x_l, y_i)\Delta t}U_H + \left(1 - \frac{h}{2b_2(x_l, y_i)\Delta t}\right)U_D, \quad (4-1-17)$$

Similarly

$$U_{R_1} \approx \frac{h}{2b_2(x_l, y_i)\Delta t}U_G + \left(1 - \frac{h}{2b_2(x_l, y_i)\Delta t}\right)U_C. \quad (4-1-18)$$

Combining the four above equations, we can get

$$\begin{aligned} U_{O_1} &\approx \frac{1}{2}U_{M_1} + \frac{1}{2}U_{N_1} \\ &\approx \frac{h}{4b_2(x_l, y_i)\Delta t}U_F + \left(\frac{1}{2} - \frac{h}{4b_2(x_l, y_i)\Delta t}\right)U_B + \frac{b_1(x_l, y_i)}{2b_2(x_l, y_i)}U_{S_1} + \left(\frac{1}{2} - \frac{b_1(x_l, y_i)}{2b_2(x_l, y_i)}\right)U_{R_1} \\ &\approx \frac{h}{4b_2(x_l, y_i)\Delta t}U_F + \left(\frac{1}{2} - \frac{h}{4b_2(x_l, y_i)\Delta t}\right)U_B + \frac{b_1(x_l, y_i)h}{4b_2(x_l, y_i)^2\Delta t}U_H + \frac{b_1(x_l, y_i)}{2b_2(x_l, y_i)}\left(1 - \frac{h}{2b_2(x_l, y_i)\Delta t}\right)U_D \\ &\quad + \left(\frac{1}{2} - \frac{b_1(x_l, y_i)}{2b_2(x_l, y_i)}\right)\frac{h}{2b_2(x_l, y_i)\Delta t}U_G + \left(\frac{1}{2} - \frac{b_1(x_l, y_i)}{2b_2(x_l, y_i)}\right)\left(1 - \frac{h}{2b_2(x_l, y_i)\Delta t}\right)U_C. \end{aligned} \quad (4-1-19)$$

According to $|FG| = h$ and (4-1-12), we have

$$|CO_1| = \sqrt{\left(\frac{h}{2}\right)^2 + \left(\frac{|R_1N_1|}{2}\right)^2 + \left(\frac{|G_1P_1|}{2}\right)^2} = \frac{h\varphi(x_l, y_i)}{2b_2(x_l, y_i)}. \quad (4-1-20)$$

Then, combining (4-1-13),(4-1-19),(4-1-20), the following holds

$$\begin{aligned}
 & \left[\varphi(x,y) \frac{\partial u}{\partial \tau(x,y)} \right]_{x=x_l, y=y_i}^{t=t^n} \approx \frac{2b_2(x_l, y_i)}{h} (U_C - U_{O_1}) \\
 & \approx \left[\frac{2b_2(x_l, y_i)}{h} - \frac{2b_2(x_l, y_i)}{h} \left(\frac{1}{2} - \frac{b_1(x_l, y_i)}{2b_2(x_l, y_i)} \right) \left(1 - \frac{h}{2b_2(x_l, y_i)\Delta t} \right) \right] u(x_l, y_i, t^n) \\
 & - \frac{1}{2\Delta t} u(x_l, y_{i-1}, t^{n-1}) - \frac{2b_2(x_l, y_i)}{h} \left(\frac{1}{2} - \frac{h}{4b_2(x_l, y_i)\Delta t} \right) u(x_l, y_{i-1}, t^n) \\
 & - \frac{b_1(x_l, y_i)}{2b_2(x_l, y_i)\Delta t} u(x_{l-1}, y_i, t^{n-1}) - \frac{b_1(x_l, y_i)}{h} \left(1 - \frac{h}{2b_2(x_l, y_i)\Delta t} \right) u(x_{l-1}, y_i, t^n) \\
 & - \frac{1}{\Delta t} \left(\frac{1}{2} - \frac{b_1(x_l, y_i)}{2b_2(x_l, y_i)} \right) u(x_l, y_i, t^{n-1}). \tag{4-1-21}
 \end{aligned}$$

The approximate method for $\nabla \cdot (a(x,y)\nabla u)_{x=x_l, y=y_i}^{t=t^n}$ is given by the method of central difference quotient, specific deduction process and results are consistent with formula (2-3-14).

For equations (4-1-21) and (2-3-14), taking the approximate equal sign as an equal sign, using $u, u_{l,i}^n, f_{l,i}^n$ replace $U, u(x_l, y_i, t^n), f(x_l, y_i, t^n)$, and bringing them into (4-1-2), then the difference format of the Case 2 can be obtained as

$$\begin{cases} \frac{2b_2(x_l, y_i)}{h} (u_C - u_{O_1}) - \delta_x(a(x, y_i)\delta_x u_i^n)_l - \delta_y(a(x_l, y)\delta_y u_l^n)_i = f_{l,i}^n, \\ \quad \quad \quad 1 \leq l, i \leq p-1, 1 \leq n \leq m, \\ u_{0,i}^n = g_0(y_i, t^n), u_{p,i}^n = g_1(y_i, t^n), \quad 1 \leq i \leq p-1, 1 \leq n \leq m, \\ u_{l,0}^n = g_2(x_l, t^n), u_{l,p}^n = g_3(x_l, t^n), \quad 0 \leq l \leq p, 1 \leq n \leq m, \\ u_{l,i}^0 = u_0(x_l, y_i), \quad 0 \leq l, i \leq p. \end{cases} \tag{4-1-22}$$

Case 3

Case 3 is shown in Figure 4-1(c), at this case $b_1(x_l, y_i), b_2(x_l, y_i)$ satisfy $b_1(x_l, y_i) > h/\Delta t$ and $b_2(x_l, y_i) < b_1(x_l, y_i)$. The characteristic line starting from point C intersects the grid plane $AEHD$ at point P_2 . Across point P_2 , we can draw a parallel line of AE and DH . The line intersects AD at point G_2 . Taking the midpoint of P_2G_2 as K_2 , acrossing K_2 , we can draw a plane parallel to the $EFGH$ as $M_2S_2R_2$. Set the characteristic line and plane $M_2S_2R_2$ intersects at point O_2 , then connects point M_2 and point O_2 and extends the line to intersect S_2R_2 at point N_2 .

We have known that $|GH| = h$ and the slope of characteristic line is $\left(\frac{b_1(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{b_2(x_l, y_i)}{\varphi(x_l, y_i)}, \frac{1}{\varphi(x_l, y_i)} \right)$, so we can get

$$|G_2P_2| = \frac{h}{b_1(x_l, y_i)}, |R_2N_2| = \frac{hb_2(x_l, y_i)}{b_1(x_l, y_i)}. \quad (4-1-23)$$

We firstly give the approximation of $\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)}$ at node (x_l, y_i, t^n)

$$\left[\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)} \right]_{x=x_l, y=y_i}^{t=t^n} \approx \varphi(x_l, y_i) \frac{U_C - U_{O_2}}{|CO_2|}. \quad (4-1-24)$$

From the fact that K_2 is the midpoint of line segment P_2G_2 , then according to geometric knowledge, point O_2 is also the midpoint of line segment M_2N_2 , so obviously

$$U_{O_2} \approx \frac{1}{2} U_{M_2} + \frac{1}{2} U_{N_2}. \quad (4-1-25)$$

The approximation of U_{M_2} can be obtained by linear interpolation of U_D and U_H ,

$$U_{M_2} \approx \frac{h}{2b_1(x_l, y_i)\Delta t} U_H + \left(1 - \frac{h}{2b_1(x_l, y_i)\Delta t} \right) U_D. \quad (4-1-26)$$

Similarly

$$U_{N_2} \approx \frac{b_2(x_l, y_i)}{b_1(x_l, y_i)} U_{S_2} + \left(1 - \frac{b_2(x_l, y_i)}{b_1(x_l, y_i)} \right) U_{R_2}. \quad (4-1-27)$$

The approximation of U_{S_2} can be obtained by linear interpolation of U_B and U_F ,

$$U_{S_2} \approx \frac{h}{2b_1(x_l, y_i)\Delta t} U_F + \left(1 - \frac{h}{2b_1(x_l, y_i)\Delta t} \right) U_B. \quad (4-1-28)$$

Similarly

$$U_{R_2} \approx \frac{h}{2b_1(x_l, y_i)\Delta t} U_G + \left(1 - \frac{h}{2b_1(x_l, y_i)\Delta t} \right) U_C. \quad (4-1-29)$$

Combining the four above equations, we can get

$$\begin{aligned} U_{O_2} &\approx \frac{1}{2} U_{M_2} + \frac{1}{2} U_{N_2} \\ &\approx \frac{h}{4b_1(x_l, y_i)\Delta t} U_H + \left(\frac{1}{2} - \frac{h}{4b_1(x_l, y_i)\Delta t} \right) U_D + \frac{b_2(x_l, y_i)}{2b_1(x_l, y_i)} U_{S_2} + \left(\frac{1}{2} - \frac{b_2(x_l, y_i)}{2b_1(x_l, y_i)} \right) U_{R_2} \\ &\approx \frac{h}{4b_1(x_l, y_i)\Delta t} U_H + \left(\frac{1}{2} - \frac{h}{4b_1(x_l, y_i)\Delta t} \right) U_D + \frac{b_2(x_l, y_i)h}{4b_1(x_l, y_i)^2\Delta t} U_F + \frac{b_2(x_l, y_i)}{2b_1(x_l, y_i)} \left(1 - \frac{h}{2b_1(x_l, y_i)\Delta t} \right) U_B \\ &\quad + \left(\frac{1}{2} - \frac{b_2(x_l, y_i)}{2b_1(x_l, y_i)} \right) \frac{h}{2b_1(x_l, y_i)\Delta t} U_G + \left(\frac{1}{2} - \frac{b_2(x_l, y_i)}{2b_1(x_l, y_i)} \right) \left(1 - \frac{h}{2b_1(x_l, y_i)\Delta t} \right) U_C. \end{aligned} \quad (4-1-30)$$

According to $|GH| = h$ and (4-1-23), we have

$$|CO_2| = \sqrt{\left(\frac{h}{2} \right)^2 + \left(\frac{|R_2N_2|}{2} \right)^2 + \left(\frac{|G_2P_2|}{2} \right)^2} = \frac{h\varphi(x_l, y_i)}{2b_1(x_l, y_i)}. \quad (4-1-31)$$

Then, combining (4-1-24), (4-1-30), (4-1-31), the following holds

$$\begin{aligned} \left[\varphi(x, y) \frac{\partial u}{\partial \tau(x, y)} \right]_{x=x_l, y=y_i}^{t=t^n} &\approx \frac{2b_1(x_l, y_i)}{h} (U_C - U_{O_2}) \\ &\approx \left[\frac{2b_1(x_l, y_i)}{h} - \frac{2b_1(x_l, y_i)}{h} \left(\frac{1}{2} - \frac{b_2(x_l, y_i)}{2b_1(x_l, y_i)} \right) \left(1 - \frac{h}{2b_1(x_l, y_i)\Delta t} \right) \right] u(x_l, y_i, t^n) \end{aligned}$$

$$\begin{aligned}
 & -\frac{1}{2\Delta t}u(x_{l-1},y_i,t^{n-1}) - \frac{2b_1(x_l,y_i)}{h}\left(\frac{1}{2} - \frac{h}{4b_1(x_l,y_i)\Delta t}\right)u(x_{l-1},y_i,t^n) \\
 & - \frac{b_2(x_l,y_i)}{2b_1(x_l,y_i)\Delta t}u(x_l,y_{i-1},t^{n-1}) - \frac{b_2(x_l,y_i)}{h}\left(1 - \frac{h}{2b_1(x_l,y_i)\Delta t}\right)u(x_l,y_{i-1},t^n) \\
 & - \frac{1}{\Delta t}\left(\frac{1}{2} - \frac{b_2(x_l,y_i)}{2b_1(x_l,y_i)}\right)u(x_l,y_i,t^{n-1}).
 \end{aligned} \tag{4-1-32}$$

The approximate method for $\nabla \cdot (a(x,y)\nabla u)_{x=x_l,y=y_i}^{t=t^n}$ is given by the method of central difference quotient, specific deduction process and results are consistent with formula (2-3-14).

For equations (4-1-32) and (2-3-14), taking the approximate equal sign as an equal sign, using $u, u_{l,i}^n, f_{l,i}^n$ replace $U, u(x_l, y_i, t^n), f(x_l, y_i, t^n)$, and bringing them into (4-1-2), then the difference format of the Case 3 can be obtained as

$$\begin{cases} \frac{2b_1(x_l,y_i)}{h}(u_c - u_{o_2}) - \delta_{\bar{x}}(a(x,y_i)\delta_x u_i^n)_l - \delta_{\bar{y}}(a(x_l,y)\delta_y u_l^n)_i = f_{l,i}^n, \\ \quad \quad \quad 1 \leq l, i \leq p-1, 1 \leq n \leq m, \\ u_{0,i}^n = g_0(y_i, t^n), u_{p,i}^n = g_1(y_i, t^n), \quad 1 \leq i \leq p-1, 1 \leq n \leq m, \\ u_{l,0}^n = g_2(x_l, t^n), u_{l,p}^n = g_3(x_l, t^n), \quad 0 \leq l \leq p, 1 \leq n \leq m, \\ u_{l,i}^0 = u_0(x_l, y_i), \quad 0 \leq l, i \leq p. \end{cases} \tag{4-1-33}$$

In summary, we have discussed the three cases when the characteristic line intersects the grid plane. Next, we will unify the above three cases into one expression.

First, let $s_1 = s_{1,l,i} = \text{sign}(b_1(x_l, y_i) - h/\Delta t)$, $s_2 = s_{2,l,i} = \text{sign}(b_2(x_l, y_i) - h/\Delta t)$, $s_3 = s_{3,l,i} = \text{sign}(b_1(x_l, y_i) - b_2(x_l, y_i))$. According to the conditions of the three cases, let $J_1 = J_{1,l,i} = \left(1 - s_1 \frac{(1+s_1)}{2}\right)\left(1 - s_2 \frac{(1+s_2)}{2}\right)$, $J_2 = J_{2,l,i} = s_2 \frac{(1+T_2)}{2}\left(1 - T_3 \frac{(1+s_3)}{2}\right)$, $J_3 = J_{3,l,i} = s_1 \frac{(1+s_1)}{2} s_3 \frac{(1+s_3)}{2}$. The explanation for J_1, J_2, J_3 is as follows: If and only if the case when characteristic line intersects the grid is case $i (i = 1, 2, 3)$, $J_i = 1$, otherwise $J_i = 0$. Then the two-dimensional improved characteristic difference scheme is as follows

$$\begin{cases} \frac{J_1(u_c - u_0)}{\Delta t} + \frac{2J_2b_2(u_c - u_{o_1})}{h} + \frac{2J_3b_1(u_c - u_{o_2})}{h} - \delta_{\bar{x}}(a(x,y_i)\delta_x u_i^n)_l - \delta_{\bar{y}}(a(x_l,y)\delta_y u_l^n)_i = f_{l,i}^n, \\ \quad \quad \quad 1 \leq l, i \leq p-1, 1 \leq n \leq m, \\ u_{0,i}^n = g_0(y_i, t^n), u_{p,i}^n = g_1(y_i, t^n), \quad 1 \leq i \leq p-1, 1 \leq n \leq m, \\ u_{l,0}^n = g_2(x_l, t^n), u_{l,p}^n = g_3(x_l, t^n), \quad 0 \leq l \leq p, 1 \leq n \leq m, \\ u_{l,i}^0 = u_0(x_l, y_i), \quad 0 \leq l, i \leq p. \end{cases} \tag{4-1-34}$$

4. 2 Numerical Experiments

For the two-dimensional linear improved characteristic difference scheme, we conducted a total of two numerical experiments. In the Experiment 1, the convergence of the two-dimensional linear improved characteristic difference scheme is verified. In the Experiment 2, the superiority of the two-dimensional linear improved characteristic difference scheme over the two-dimensional linear characteristic difference scheme is verified.

4. 2. 1 Experiment 1

For two-dimensional problems, we select the following equation to verify the convergence of the improved characteristic difference scheme

$$\begin{cases} \frac{\partial u}{\partial t} + b_1 \frac{\partial u}{\partial x} + b_2 \frac{\partial u}{\partial y} - \nabla \cdot (a \nabla u) = f(x, y, t), & (x, y) \in (0, L) \times (0, L), t \in (0, T], \\ u(0, y, t) = g_0(y, t), u(1, y, t) = g_1(y, t), & y \in (0, L), t \in (0, T], \\ u(x, 0, t) = g_2(x, t), u(x, 1, t) = g_3(y, t), & x \in [0, L], t \in (0, T], \\ u(x, y, 0) = u_0(x, y), & (x, y) \in [0, L] \times [0, L]. \end{cases} \quad (4-2-1)$$

Take $b_1 = 1, b_2 = 3$ and $a = 0.05, a = 0.005$ separately, and set $L = 1, T = 0.5$. Suppose the analytical solution of the equation is $u(x, y, t) = e^{-t} \sin(\pi x) \sin(\pi y)$, then the initial boundary value conditions of the equation and the right term $f(x, y, t)$ can be determined by the analytical solution and formula (4-2-1), the specific expression is omitted here.

Based on the above conditions, the calculation results by improved characteristic difference scheme are shown in Table 4-1.

Table 4-1

a	h	Δt	$E_{\infty}(h, \Delta t)$	Order
0.05	1/20	1/40	0.1332	*
	1/40	1/80	0.0708	1.88
	1/80	1/160	0.0365	1.94
	1/160	1/320	0.0186	1.96
0.005	1/20	1/40	0.1494	*
	1/40	1/80	0.0791	1.89
	1/80	1/160	0.0409	1.93
	1/160	1/320	0.0208	1.97

where

$$E_{\infty}(h, \Delta t) = \max_{\substack{0 \leq l, i \leq p \\ 0 \leq n \leq m}} |u(x_{l,i}, t^n) - u_{l,i}^n|, \text{ Order} = E_{\infty}(2h, 2\Delta t) / E_{\infty}(h, \Delta t).$$

According to the above Table 4-1, we can see that when the time step and space step are reduced to 1/2 of the original, the maximum error is also reduced to about half of the original. Thus, we numerically verify the two-dimensional linear improved characteristic difference scheme is convergent, and its error order should be the same order as $O(h + \Delta t)$. At the same time, the scheme still has good convergence when deals with the case with small diffusion term coefficient.

4.2.2 Experiment 2

In order to further verify the superiority of the two-dimensional linear improved characteristic difference scheme over the two-dimensional linear characteristic difference scheme numerically, let us consider the following equation

$$\begin{cases} \frac{\partial u}{\partial t} + b_1 \frac{\partial u}{\partial x} + b_2 \frac{\partial u}{\partial y} - \nabla \cdot (a \nabla u) = f(x, y, t), & (x, y) \in (0, L) \times (0, L), t \in (0, T], \\ u(0, y, t) = g_0(y, t), u(1, y, t) = g_1(y, t), & y \in (0, L), t \in (0, T], \\ u(x, 0, t) = g_2(x, t), u(x, 1, t) = g_3(y, t), & x \in [0, L], t \in (0, T], \\ u(x, y, 0) = u_0(x, y), & (x, y) \in [0, L] \times [0, L]. \end{cases} \quad (4-2-2)$$

Suppose the analytical solution of the equation is $u(x, y, t) = \frac{1}{2} e^{\frac{1}{2}(x+y)-t}$, then the initial boundary value conditions of the equation and the right term $f(x, y, t)$ can be determined by the analytical solution and formula (4-2-2), the specific expression is omitted here.

Set $L = 1, T = 1, h = 0.01$ and $\Delta t = 0.005$. By taking different a, b_1, b_2 we can get Table 4-2.

Table 4-2

a	(b_1, b_2)	$E^1_{\infty}(h, \Delta t)$	$E^2_{\infty}(h, \Delta t)$
0.01	(1, 1)	0.0044	0.0044
	(5, 5)	0.0032	0.0012
	(10, 10)	0.0064	0.0010
0.001	(1, 1)	0.0010	0.0010
	(5, 5)	0.0028	0.0007
	(10, 10)	0.0064	0.0007

where

$$E_{\infty}^i(h, \Delta t) = \max_{\substack{0 \leq l, i \leq p \\ n=m}} |u^i(x_l, y_i, t^n) - u_{l,i}^{i^n}|, (i = 1, 2).$$

$i = 1, 2$ represents characteristic difference scheme and improved characteristic difference scheme respectively.

From the above table, we can see that when (b_1, b_2) is small, whether it is the characteristic difference scheme or the improved characteristic difference scheme, they both have a good fitting effect. But when the diffusion term coefficient a is taken, for the larger convection term coefficients (b_1, b_2) , the error calculated using the two-dimensional improved characteristic difference scheme is obviously smaller than the error obtained by the two-dimensional characteristic difference scheme. The conclusion of this numerical experiment is consistent with the experiments conducted in one-dimensional problems.

Conclusion

This paper first introduces the one-dimensional characteristic difference scheme in detail, gives the complete error analysis process of the one-dimensional linear characteristic difference scheme in the case of constant coefficients, and extends the one-dimensional characteristic difference scheme to the two-dimensional linear case. Then, based on the analysis of the characteristic difference scheme, we found that the traditional characteristic difference scheme is prone to large numerical diffusion when the convection term coefficient is large. In order to eliminate this numerical phenomenon, this paper constructs a one-dimensional linear improved characteristic difference scheme and give its error analysis in the case of constant coefficients, and further extend it to the case of one-dimensional nonlinearity and two-dimensional linearity. In this paper, the problem of one-dimensional linearity, one-dimensional nonlinearity, and two-dimensional linearity numerical examples are given for the improved characteristic difference scheme, and numerical experiments are carried out on these examples using the software MATLAB. Error analysis and numerical results show that the improved characteristic difference scheme we constructed is effective and accurate. And this scheme can also avoid the serious numerical diffusion phenomenon due to large convection term coefficient. At the same time, due to the characteristics of the improved characteristic difference scheme, compared with the characteristic difference scheme, the improved scheme can also be better when solve nonlinear problems such as Burgers equation.

However, since the author's theoretical level and the limited length of the article, this article needs to be further given for the following work: First, the error analysis of the improved characteristic difference scheme in one-dimensional nonlinearity and two-dimensional linear cases needs to be further given; It is also necessary to try to further extend the improved characteristic difference scheme to the practical application model, to further test the practicability of the improved characteristic difference scheme, and exert its corresponding actual value.

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